A.1 Kinematics

The important stuff, then worked examples. Every idea explained from the ground up.

WHAT THIS TOPIC IS ABOUT Kinematics is just the **description of motion**: where is something, how fast is it going, and how quickly is that speed changing? It does **not** ask *why* it moves like that (that is forces, the next topic). Everything in this guide grows out of one chain of ideas: **position → velocity → acceleration**.

How this guide is organised

Chapter	What is in it	How to use it
1. The Important Stuff	Each idea in plain words first, then the equation, then what it means. Numbered equations, labelled figures, and tables.	Read in order the first time. Come back to single sections when you are stuck.
2. Worked Examples	22 solved problems, grouped by topic, all solved with the **same six-step method**.	Cover the solution, try the question, then compare. Look at the diagram first.
3. Quick Reference	All formulas, a "which method?" guide, a table of common mistakes and a self-check list.	Use for revision and before tests.

The six-step method used in every worked example

Step	What you do	Why
1. Draw	Sketch the situation. Choose the origin and which direction is **positive**.	Signs are the biggest source of lost marks. Decide them once, at the start.
2. List	Write what you **know** and what you **want**, with units.	Shows you which quantities are missing.
3. Choose	Pick the equation (or graph rule) that contains what you want and **leaves out** the quantity you do not have.	Avoids needless extra calculations.
4. Rearrange	Solve for the unknown **in symbols first**.	Fewer rounding errors; easier to check.
5. Substitute	Put in numbers **with units** and calculate.	Wrong units show up as wrong answers.
6. Check	Is the sign right? Is the size sensible? Does it match a graph or a limiting case?	Catches slips before the examiner does.

Unless a question says otherwise: take $\mathbf{g = 9.81\ m\ s^{-2}}$ (the value in the IB data booklet) and **ignore air resistance**.

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Chapter 1

The Important Stuff

1.1 The big picture: how everything connects

Kinematics looks like a long list of topics: graphs, equations, free fall, projectiles. They are all **one idea seen from different angles**. Three quantities describe motion, and each one is *how quickly the previous one changes*:

- **Position** (s) tells you *where* an object is.
- **Velocity** (v) tells you *how quickly position is changing*, and in which direction.
- **Acceleration** (a) tells you *how quickly velocity is changing*.

On a graph, "how quickly something changes" is a **gradient** (slope), and going the other way (adding up all the little changes) is an **area**. That is why graphs are so central: they are the picture of this chain.

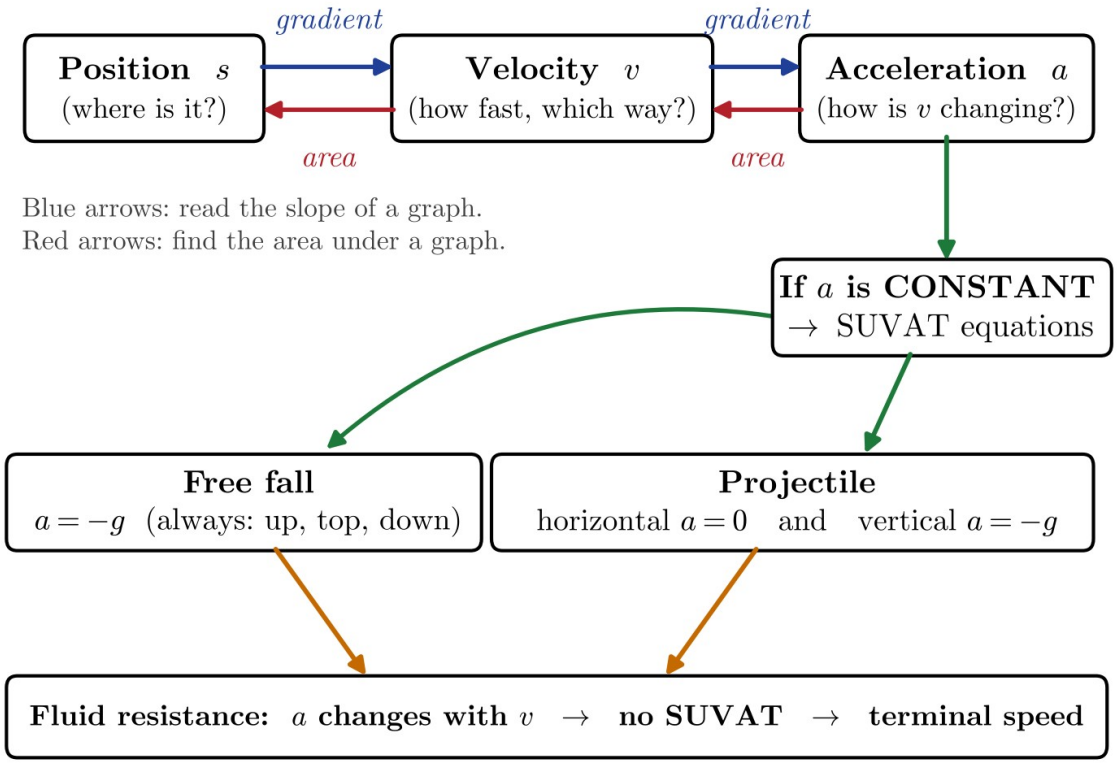


Figure 1.1: How the ideas in kinematics connect. Gradients take you along the chain from position to velocity to acceleration; areas take you back. Constant acceleration gives SUVAT; free fall and projectiles are special cases; fluid resistance breaks the "constant acceleration" assumption.

IN PLAIN ENGLISH Velocity answers "how fast is my position changing?" Acceleration answers "how fast is my velocity changing?" If you can say those two sentences without looking, you already understand most of this topic. Everything else is learning to **read** that idea from a graph, an equation or a diagram.

1.2 Distance, displacement and signs

Position is always measured **relative to a chosen origin**. You choose the origin and which direction counts as positive; the physics does not care, but you must stay consistent.

Distance is the total length of the path travelled. It only ever adds up, so it is never negative. **Displacement** is the straight-line change in position from start to finish, *with direction*:

$$\Delta s = s_{final} - s_{initial} \tag{1.1}$$

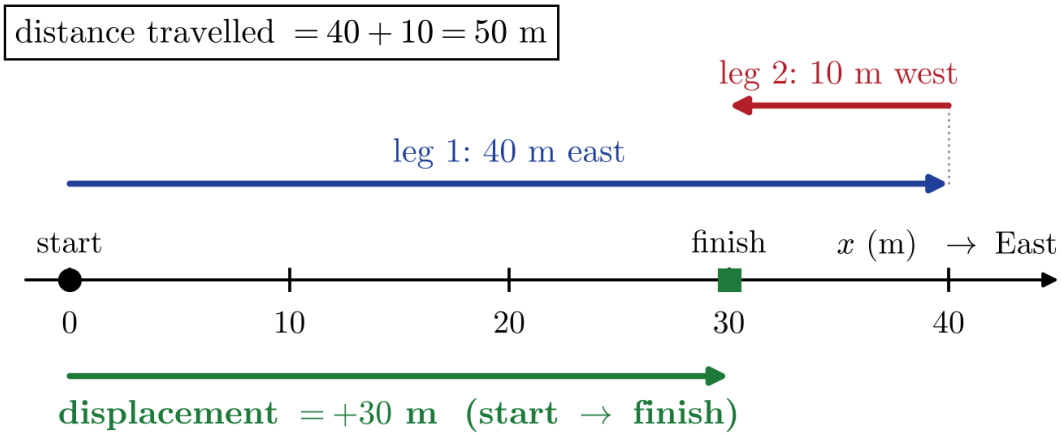


Figure 1.2: A walk of 40 m east followed by 10 m west. The distance travelled is 50 m, but the displacement is only +30 m (30 m east of the start).

Quantities that have only a size are **scalars**. Quantities that also have a direction are **vectors**. In one dimension, direction is shown by a **sign**: after you choose "east is positive", a displacement of −5 m simply means 5 m to the west.

Scalar (size only)	**Vector (size and direction)**	**Link**
distance (m)	displacement (m)	displacement is distance *with direction*, measured start → finish
speed (m s^{-1})	velocity (m s^{-1})	velocity is displacement per unit time
time (s)	acceleration (m s^{-2})	acceleration is change of velocity per unit time

WATCH OUT A **negative** sign means "opposite to the positive direction". It does not mean "smaller" or "wrong". A velocity of −8 m s^{-1} is *faster* than +3 m s^{-1}. Also: the distance travelled is always **at least as big** as the size of the displacement. They are equal only for straight-line motion that never turns back.

1.3 Speed and velocity

Speed is how much distance is covered per second. **Velocity** is how much *displacement* is covered per second, so velocity has a direction. For a whole journey we use averages:

$$\text{average speed} = \text{total distance} \,/\, \text{total time} \tag{1.2}$$

$$\text{average velocity } \bar{v} = \Delta s \,/\, \Delta t = (s_2 - s_1) \,/\, (t_2 - t_1) \tag{1.3}$$

An **average** hides what happens in between. The **instantaneous velocity** is the velocity at one moment. It is what you get from the average when you make the time interval Δt shrink towards zero:

$$v = ds/dt \quad \text{(the gradient of the tangent to the s–t graph)} \tag{1.4}$$

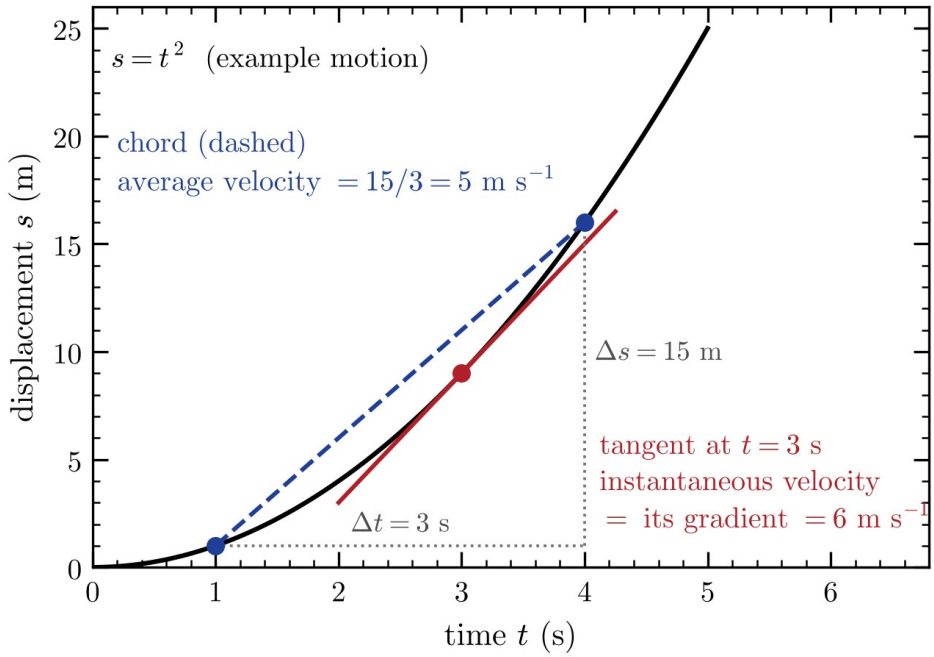


Figure 1.3: On a displacement–time graph, the **chord** between two points (dashed) gives the *average* velocity over that interval. The **tangent** at a single point (red) gives the *instantaneous* velocity at that moment.

	Average velocity	**Instantaneous velocity**
Over what?	a time interval (from t_1 to t_2)	a single instant
From an s–t graph	gradient of the **chord** joining two points	gradient of the **tangent** at that point
Calculation	Δs / Δt	draw tangent, then rise / run
Speedometer shows...	—	(the size of) the instantaneous velocity

WATCH OUT **Average speed is not the average of two speeds.** If you drive 20 km h^{-1} uphill and 30 km h^{-1} downhill over the same distance d, you spend more *time* at the slow speed. Total time = d/20 + d/30 = d/12, so the average speed is 2d ÷ (d/12) = **24 km h^{-1}**, not 25. Always use total distance ÷ total time.

Handy conversion: to go from m s^{-1} to km h^{-1} multiply by **3.6**; to go back, divide by 3.6.

1.4 Acceleration

Acceleration is the rate at which **velocity** changes. If a car's velocity rises from 4 m s^{-1} to 10 m s^{-1} in 3 seconds, it gains 2 m s^{-1} every second, so its acceleration is 2 m s^{-2} (read "2 metres per second, per second").

$$a = \Delta v \,/\, \Delta t = (v - u) \,/\, t \tag{1.5}$$

Here **u** is the *initial* velocity and **v** the *final* velocity. Acceleration is a **vector**, so it also has a sign. The key question is not "is a positive or negative?" but **"do v and a point the same way?"**

Velocity v	Acceleration a	What happens to the speed?	Example
+	+	**speeds up** (same direction)	car pulling away to the right
+	−	**slows down** (opposite directions)	car braking while moving right
−	−	**speeds up** (same direction)	ball falling when "up" is positive
−	+	**slows down** (opposite directions)	ball rising when "up" is negative

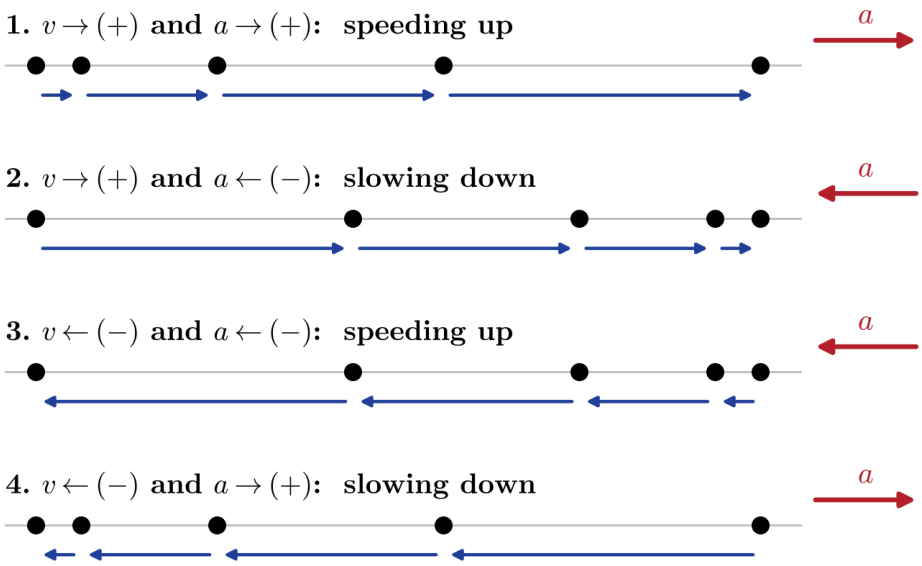


Figure 1.4: Motion diagrams. Each dot is the object's position at equal time intervals; the blue arrows show how far it moved in each interval (its velocity). When the gaps grow the object speeds up; when they shrink it slows down. The red arrow shows the direction of the acceleration.

KEY IDEA **Same direction → speeding up. Opposite directions → slowing down.** The word "deceleration" just means the acceleration points opposite to the velocity. Acceleration can also be non-zero while velocity is momentarily **zero** (for example, at the top of a throw): what matters is that the velocity is *changing*.

1.5 Motion graphs: gradients and areas

There are three graphs for the same motion. Each one is read in two ways: by its **gradient** and by the **area** under it.

Graph	Gradient (slope) gives...	Area under the graph gives...
Displacement–time (s–t)	**velocity**	(nothing useful)
Velocity–time (v–t)	**acceleration**	**displacement**
Acceleration–time (a–t)	(not needed at SL)	**change in velocity** Δv

The three graphs for one journey are shown in Figure 1.5. The object speeds up (A), cruises (B), then slows, stops and reverses (C). Notice how each graph is the slope of the one above it.

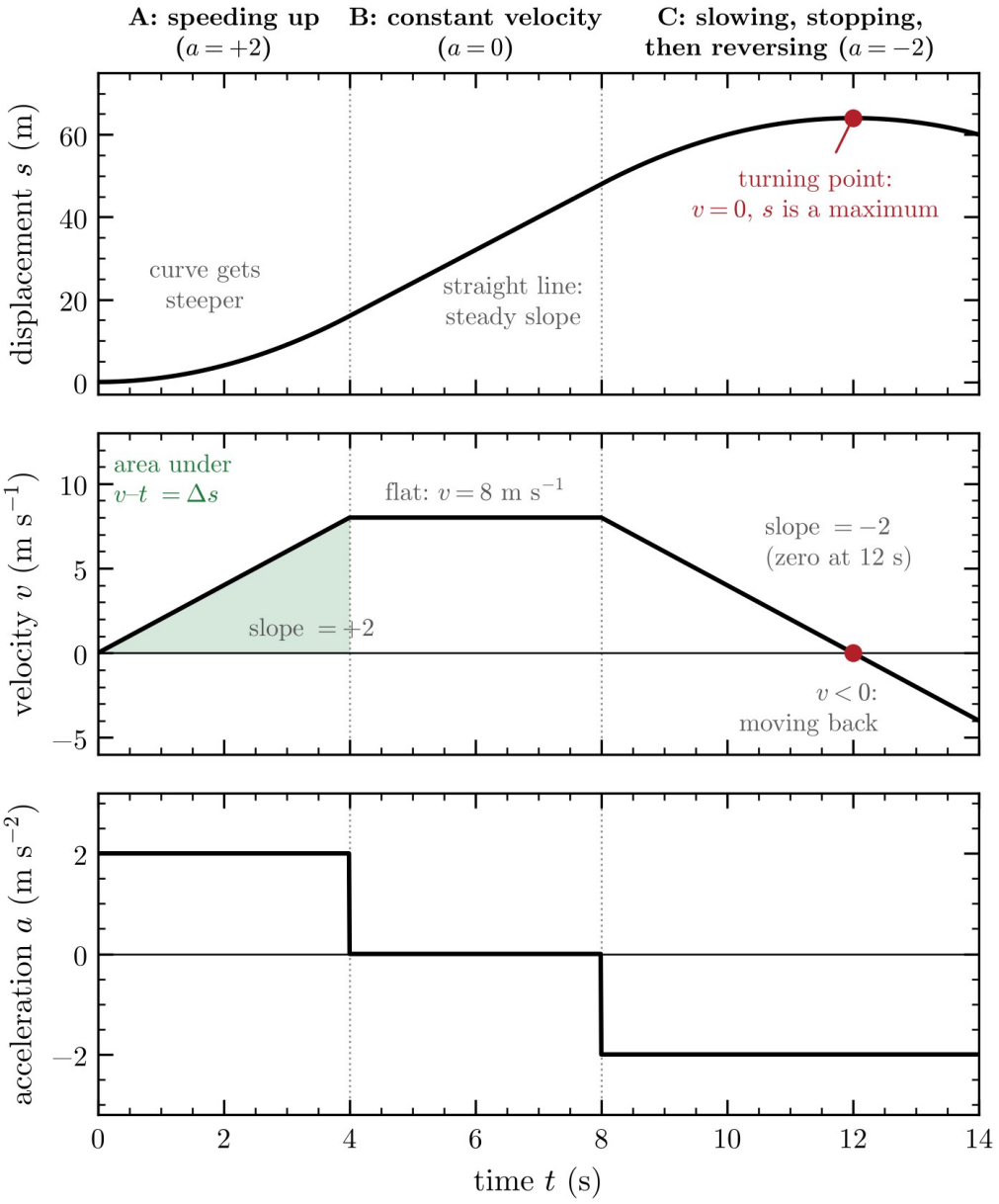


Figure 1.5: The same journey on all three graphs, sharing one time axis. The slope of the s–t graph is the height of the v–t graph; the slope of the v–t graph is the height of the a–t graph. At t = 12 s the velocity is zero, so the displacement is at its maximum: that is the turning point.

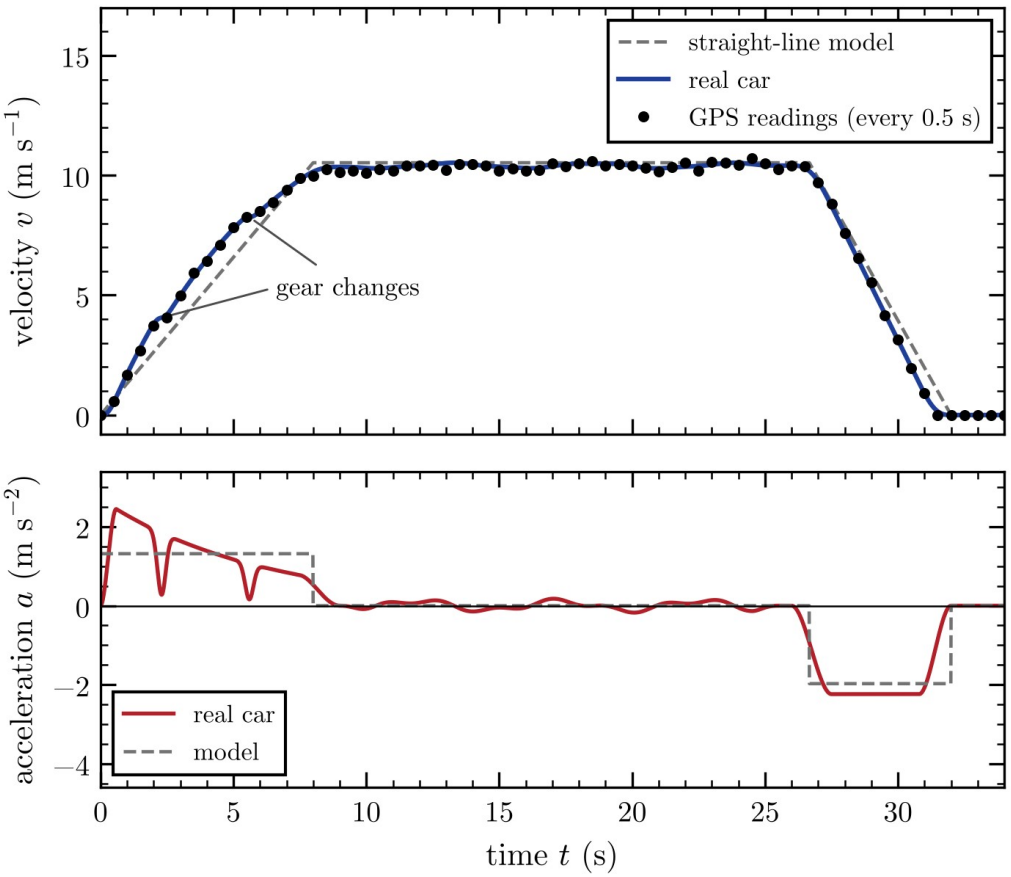


Figure 1.6: A real car pulling away from one set of traffic lights, cruising and braking at the next (computer model, with simulated GPS readings every 0.5 s). A real acceleration cannot jump from one value to another, so the real v–t graph has no sharp corners, and each gear change shows up as a brief dip in a. The straight-line model used in exam questions has the same top speed and the same area under it (about 267 m).

What each shape means

Graph	Shape	Meaning
s–t	horizontal line	at rest (gradient = 0)
s–t	straight sloped line	constant velocity
s–t	curve getting steeper / flatter	speeding up / slowing down
v–t	horizontal line	constant velocity, a = 0
v–t	straight sloped line	constant acceleration
v–t	line crossing the time axis	velocity changes sign: the object **turns round**
a–t	horizontal line (not on the axis)	constant acceleration, so SUVAT works

Using areas on a v–t graph

Areas **above** the time axis are positive displacement; areas **below** it are negative displacement. This gives a two-line rule that comes up in nearly every graph question:

- **Displacement** = the *signed* sum of the areas (add the areas above, subtract the areas below).
- **Distance travelled** = the sum of the *sizes* of all the areas (treat every area as positive).

WATCH OUT (the "hill illusion") **Do not read a graph like a picture of the path.** A v–t graph that slopes downwards does not mean the object is going down a hill; it means its velocity is decreasing. Always say what the *axes* are, then describe what the *line* does. Also: a v–t graph with a negative gradient does not always mean "slowing down": if the velocity is already negative, a negative acceleration makes it speed up in the negative direction (see the table in 1.4).

1.6 Equations of motion (SUVAT)

When the acceleration is **constant**, motion in a straight line can be described exactly by four equations. They are called the SUVAT equations after the five quantities in them:

Symbol	Quantity	Unit
s	displacement (change in position)	m
u	initial velocity (at the start of the interval)	m s^{-1}
v	final velocity (at the end of the interval)	m s^{-1}
a	acceleration (constant)	m s^{-2}
t	time taken	s

Where they come from (no calculus needed)

Take the v–t graph of constant acceleration: a straight line from u at time 0 to v at time t (Figure 1.7). We only need the two graph rules: *gradient = acceleration* and *area = displacement*.

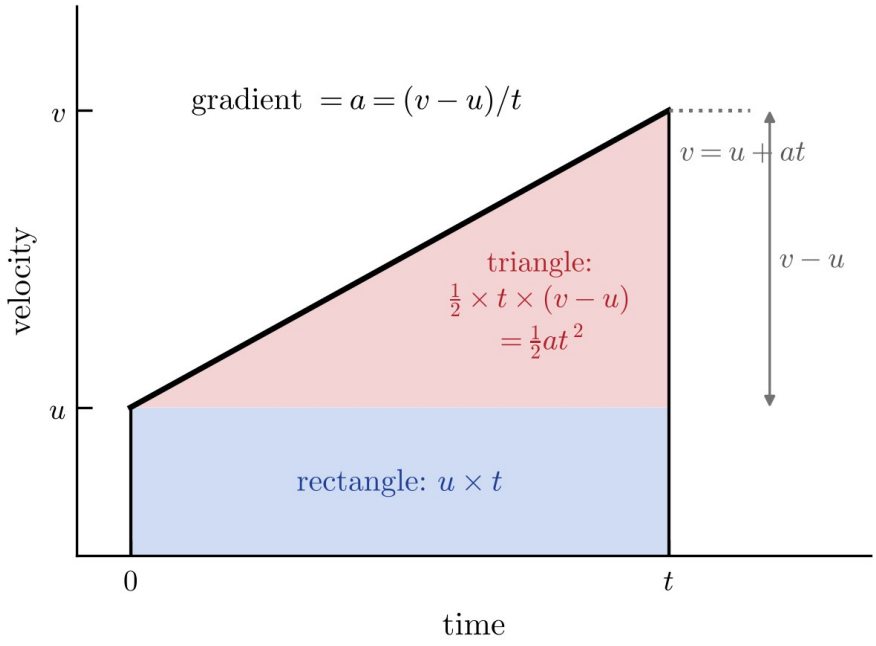


Figure 1.7: A straight-line v–t graph for constant acceleration. The area under it (a rectangle plus a triangle) is the displacement.

- **Gradient.** The gradient is a = (v − u)/t. Rearranging gives **v = u + at**.
- **Trapezium.** The area is $\frac{1}{2}$ × (sum of parallel sides) × width = $\frac{1}{2}$(u + v)t, so **s = $\frac{1}{2}$(u + v)t**.

- **Rectangle + triangle.** Rectangle: $u \times t$. Triangle: $\frac{1}{2} \times t \times (v - u) = \frac{1}{2} \times t \times at = \frac{1}{2}at^2$. So $\mathbf{s = ut + \frac{1}{2}at^2}$.
- **Eliminate t.** Put $t = (v - u)/a$ into $s = \frac{1}{2}(u + v)t$ to get $s = (v^2 - u^2)/2a$. So $\mathbf{v^2 = u^2 + 2as}$.

$$v = u + at \tag{1.6}$$

$$s = \tfrac{1}{2}(u + v)t \tag{1.7}$$

$$s = ut + \tfrac{1}{2}at^2 \tag{1.8}$$

$$v^2 = u^2 + 2as \tag{1.9}$$

WHY IT WORKS The middle equation says *displacement = average velocity × time*. For constantly changing velocity the average is exactly halfway between u and v, which is why the $\frac{1}{2}(u + v)$ appears. Every equation here is the same graph, read a different way.

Choosing the right equation

Do not pick the equation that "looks right". Write the five SUVAT quantities as a list, mark the three you **know** and the one you **want**. The fifth quantity is the one you neither know nor want: choose the equation that **leaves it out**.

Equation	Missing variable	Use it when you have no...
$v = u + at$	**s**	displacement
$s = \frac{1}{2}(u + v)t$	**a**	acceleration
$s = ut + \frac{1}{2}at^2$	**v**	final velocity
$v^2 = u^2 + 2as$	**t**	time

WATCH OUT SUVAT works **only** for constant acceleration in a straight line. If the acceleration changes (for example, a skydiver with air resistance), split the motion into stages or use graphs. For a motion in **two stages**, the *final* velocity of stage 1 is the *initial* velocity of stage 2. Quadratic equations in t (from $s = ut + \frac{1}{2}at^2$) give **two** answers: reject any that are physically impossible, such as a negative time.

EXAM TIP For "derive" questions, start by stating the assumption: *constant acceleration*. Then use the trapezium: $\mathbf{v = u + at}$ and $\mathbf{s = \frac{1}{2}(u + v)t}$ are the two shortest steps to any of the others.

1.7 Free fall

An object is in **free fall** when gravity is the *only* force acting on it. Near the Earth's surface, every object in free fall has the same acceleration, whatever its mass:

$$a = g = 9.81 \text{ m s}^{-2} \text{ directed downwards} \tag{1.10}$$

"Free fall" does **not** mean "falling". A ball thrown upward is in free fall from the moment it leaves your hand: on the way up, at the top, and on the way down. The acceleration is **always** 9.81 m s^{-2} downward. If you choose *up* as positive, then $\mathbf{a = -9.81 \text{ m s}^{-2}}$ **throughout**, and the SUVAT equations apply.

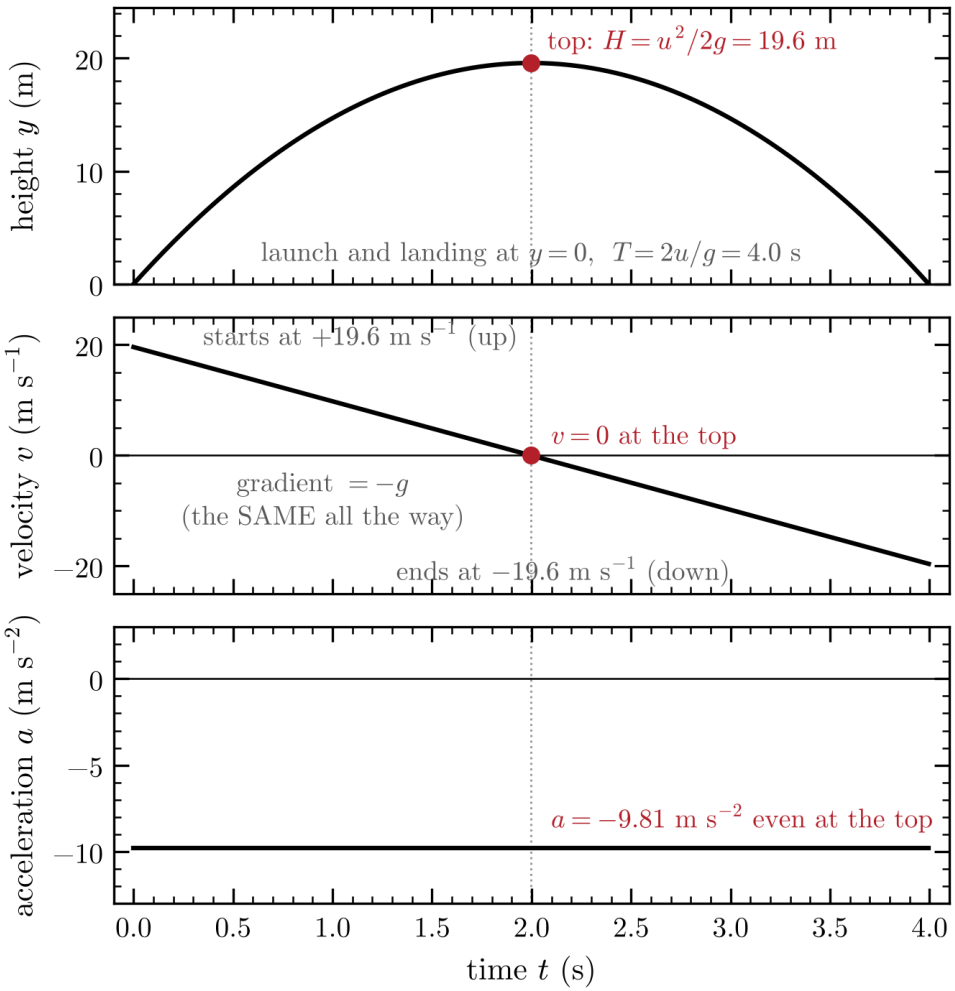


Figure 1.8: A ball thrown upward at 19.6 m s^{-1} (up is positive). The velocity decreases by 9.81 m s^{-1} every second, passes through zero at the top, and keeps going into negative values. The acceleration never changes.

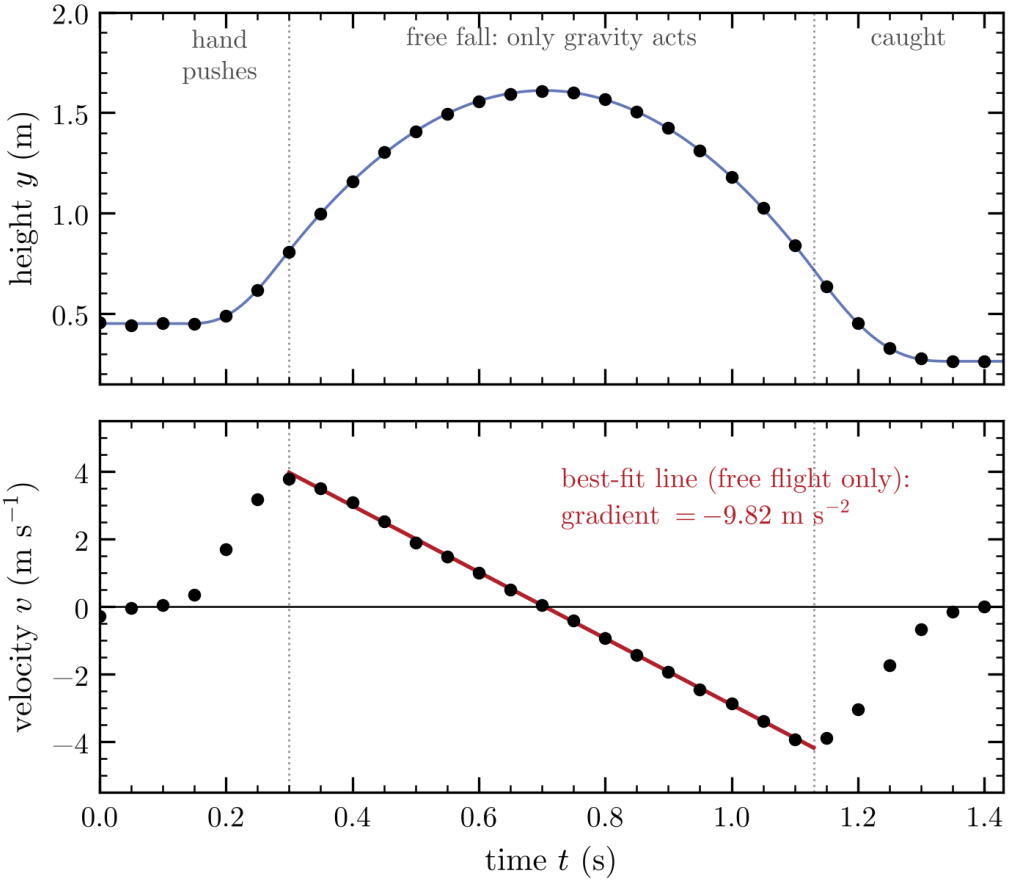


Figure 1.9: Simulated motion-sensor readings (every 0.05 s) for a ball tossed upward and caught. The ball is in free fall only between release and catch: that part of the v–t graph is a straight line with gradient -9.82 m s^{-2}, very close to −g. While the hand is pushing or catching, it is not free fall.

Useful results (launch and landing at the same height, no air resistance)

Quantity	**Result**	**How it comes from SUVAT**
time to the top	$t_{up} = u / g$	$v = u + at$ with $v = 0$
maximum height	$H = u^2 / 2g$	$v^2 = u^2 + 2as$ with $v = 0$
total time in the air	$T = 2u / g$	going up and coming down take equal times
speed on returning to the launch height	$= u$ (but downward)	the motion is symmetric
dropped from rest through height h	$t = \sqrt{(2h/g)}$, $v = \sqrt{(2gh)}$	$s = \frac{1}{2}at^2$ and $v^2 = 2as$ with $u = 0$

WHY a ≠ 0 AT THE TOP At the very top the velocity is zero **for an instant**, but the acceleration is not zero. If it were, the ball would stay at rest there forever, like a ball with no forces on it. Gravity is still pulling, which is why the velocity goes from positive (rising) to negative (falling) rather than staying at zero.

WATCH OUT The number g is *positive* (9.81). The minus sign belongs to your **sign convention** (up positive means $a = -g$). If you choose *down* as positive instead, then $a = +g$. Pick one convention for a problem and use it for *every* quantity: s, u, v and a.

1.8 Vectors: splitting into components

A velocity at an angle can be split into two perpendicular parts that act **independently**: one horizontal, one vertical. This is the tool that turns a curved, two-dimensional motion into two simple one-dimensional motions.

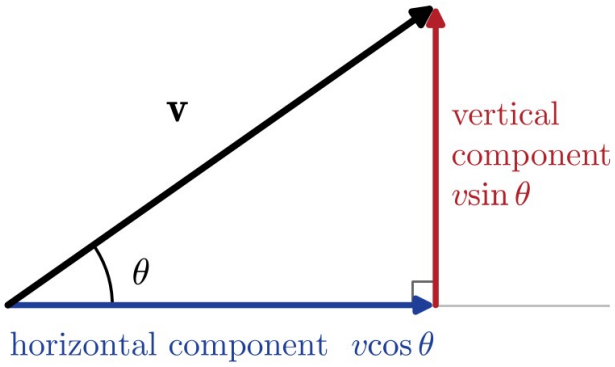


Figure 1.10: Resolving a vector of magnitude v at angle ϑ above the horizontal into a horizontal component v cos ϑ and a vertical component v sin ϑ.

$$v_x = v \cos \vartheta \tag{1.11}$$

$$v_y = v \sin \vartheta \tag{1.12}$$

To go back from components to a single vector, use Pythagoras for the size and the tangent for the angle:

$$v = \sqrt{(v_x^{\,2} + v_y^{\,2})} \tag{1.13}$$

$$\tan \vartheta = |v_y| \,/\, |v_x| \tag{1.14}$$

WATCH OUT **Check which angle you are given.** If ϑ is measured from the *horizontal*: horizontal part = cos ϑ, vertical part = sin ϑ. If ϑ is measured from the *vertical*, they swap. A quick check: when ϑ is small (almost horizontal), the horizontal component should be nearly the whole vector.

Direction of a final answer: use the *sizes* of the components in $\tan \vartheta = |v_y| \,/\, |v_x|$ and then describe it in words, for example "58° below the horizontal". This avoids the calculator giving an angle in the wrong quadrant.

1.9 Projectile motion

A **projectile** is an object that, once launched, moves under gravity alone (ignoring air resistance). Its path is a curve called a **parabola**. Two ideas unlock every projectile problem:

KEY IDEA **1. Horizontal and vertical motion are independent.** Nothing pushes sideways ($a_x = 0$), so the horizontal velocity never changes. Gravity acts only vertically ($a_y = -g$).

2. They share the same clock. The time of flight is the **link** between the two directions.

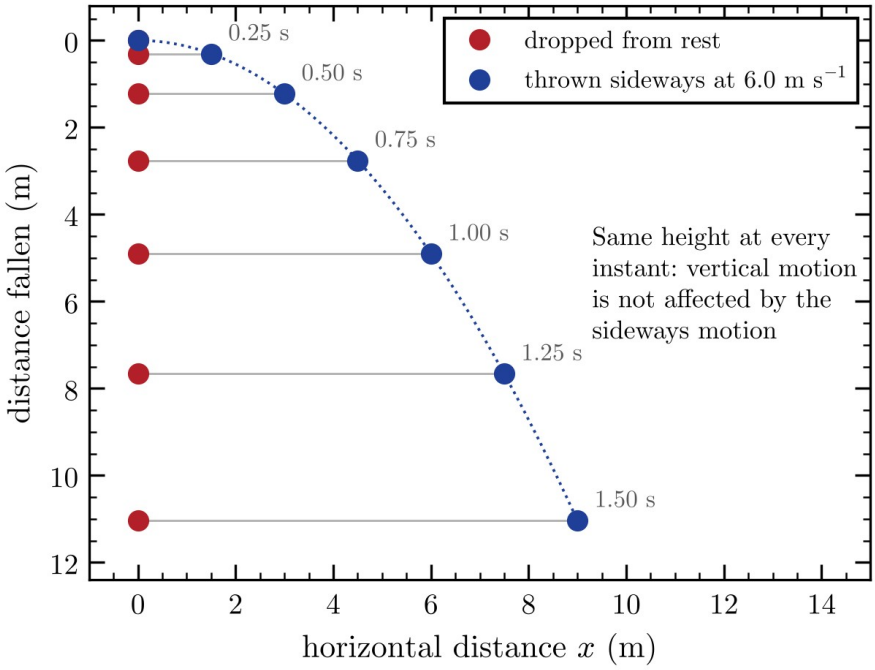


Figure 1.11: One ball is dropped; another is thrown sideways at the same moment. Positions are shown every 0.25 s. At every instant both are at the same height: the sideways motion has no effect on the fall.

	Horizontal (x)	**Vertical (y, up positive)**
Acceleration	0	$-g = -9.81$ m s^{-2}
Initial velocity	$u_x = u\cos\vartheta$	$u_y = u\sin\vartheta$
Velocity at time t	$v_x = u_x$ (constant)	$v_y = u_y - gt$
Displacement at time t	$s_x = u_x\, t$	$s_y = u_y\, t - \frac{1}{2}gt^2$
No-time equation	—	$v_y^{\,2} = u_y^{\,2} - 2g\, s_y$

$$s_x = u_x\, t \tag{1.15}$$

$$v_y = u_y - gt \tag{1.16}$$

$$s_y = u_y\, t - \tfrac{1}{2}gt^2 \tag{1.17}$$

$$v_y^{\,2} = u_y^{\,2} - 2g\, s_y \tag{1.18}$$

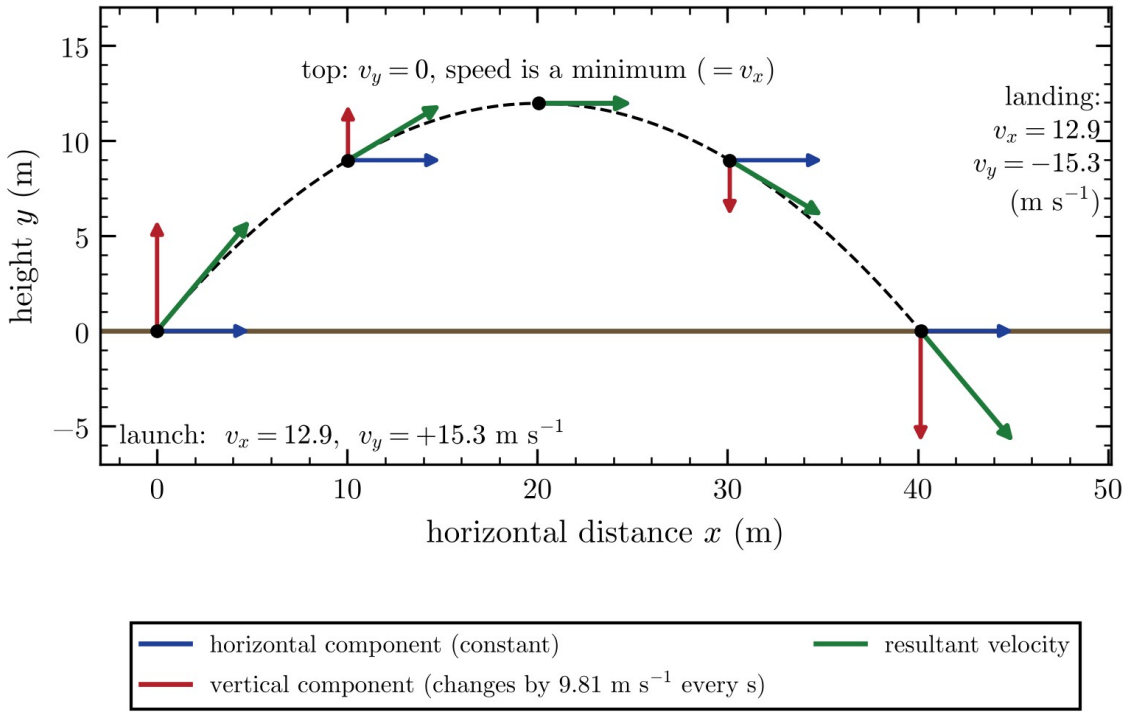


Figure 1.12: Velocity of a projectile launched at 20 m s^{-1}, 50° above the horizontal, shown at equal time steps. The blue horizontal component never changes; the red vertical component shrinks to zero at the top, then grows downward.

What is special about the top of the path?

- The **vertical** velocity is zero ($v_y = 0$), but the object is **not at rest**.
- The **speed is at its smallest**, equal to the horizontal velocity u_x.
- The acceleration is still **g downward**.
- If the object is launched horizontally ($\vartheta = 0$), $u_y = 0$: the starting point *is* the top.

The method for any projectile problem

1. **Resolve** the launch velocity into u_x and u_y.
2. Use the **vertical** equations to find the **time** (from a height, a landing level, or $v_y = 0$).
3. Put that time into the **horizontal** equation to find the distance.
4. Combine v_x and v_y with Pythagoras and $\tan\vartheta$ for the final speed and direction.

Shortcuts when the landing height equals the launch height

If the projectile lands at the *same* height it started from, and there is no air resistance, the flight is symmetric. Setting $s_y = 0$ in Eq. 1.17 and $v_y = 0$ in Eq. 1.16 gives:

$$T = 2u \sin\vartheta \,/\, g \tag{1.19}$$

$$H = (u \sin\vartheta)^2 \,/\, 2g \tag{1.20}$$

$$R = u^2 \sin 2\vartheta \,/\, g \tag{1.21}$$

Here T is the time of flight, H the maximum height above the launch point and R the horizontal range ($R = u_x\, T$, and $2 \sin\vartheta \cos\vartheta = \sin 2\vartheta$). Figure 1.13 shows what these mean.

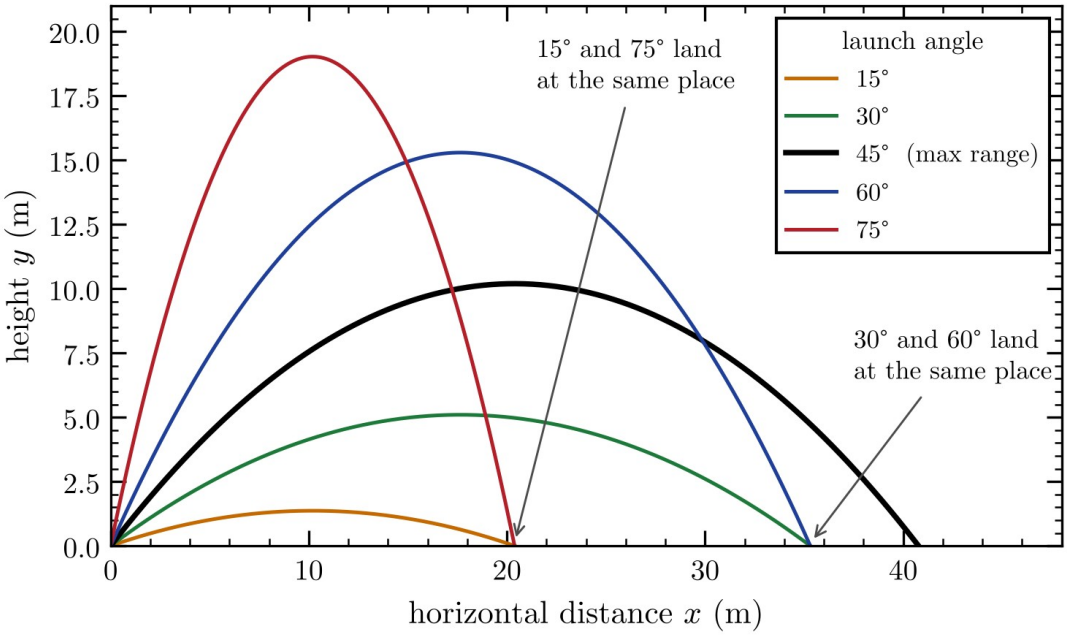


Figure 1.13: Paths for the same launch speed (20 m s^{-1}) at five angles. The range is greatest at 45°. Angles that add to 90° (30° and 60°; 15° and 75°) give the **same** range but different heights and times.

Landing at a different height

If the object lands higher or lower than the launch point, Eqs. 1.19–1.21 **do not apply**. Solve $s_y = u_y t - \frac{1}{2}gt^2$ with the correct s_y (negative if it lands lower). This is a quadratic equation in t: there will be two roots, and you keep the positive, physically sensible one.

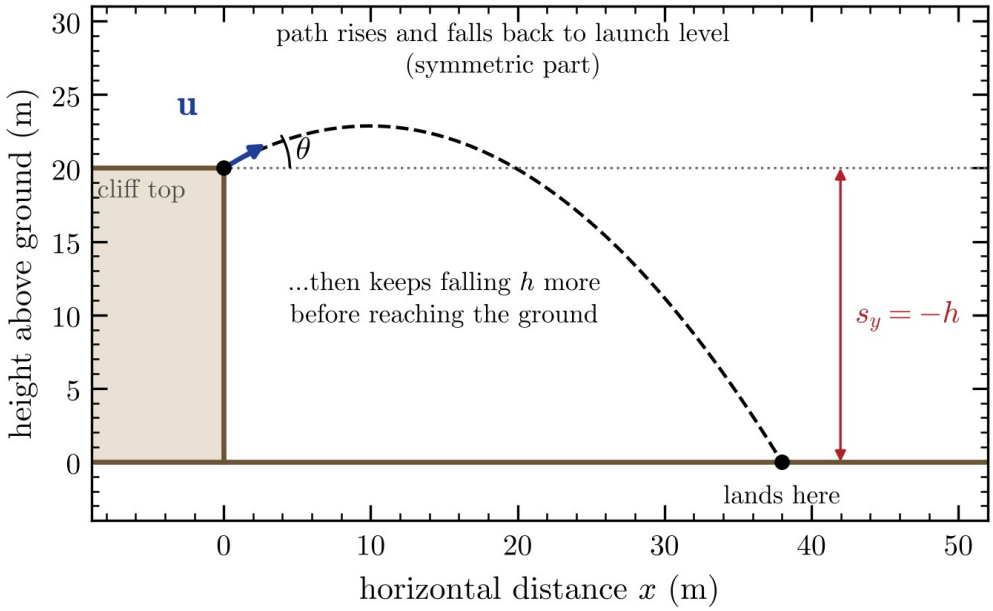


Figure 1.14: A launch from the top of a cliff. The path first rises, then falls back to the launch level, then keeps falling a further height h. Here $s_y = -h$ at the moment of landing.

WATCH OUT The formulas T, H and R are *shortcuts with conditions*: **(i)** same launch and landing height, **(ii)** no air resistance. Using them for a cliff launch, or a ball hit from above the ground, is one of the most common ways to lose marks. When in doubt, go back to Eqs. 1.15–1.18.

EXAM TIP State your assumptions: "neglect air resistance, so $a_x = 0$ and $a_y = -g$." Then show the vertical equation that gives the time **before** you move to the horizontal.

1.10 Fluid resistance and terminal speed

Everything so far ignored air. In reality, anything moving through a **fluid** (air or a liquid) feels a force, called **fluid resistance** or **drag**, that opposes its motion. Drag has four features worth knowing:

- It acts **opposite to the velocity**.
- It **increases with speed**: the faster you go, the harder the fluid pushes back.
- It is larger for a **larger cross-sectional area** and for a less streamlined shape.
- It is larger in a **denser** fluid (water pushes back far more than air).

For an object falling straight down, two forces act: the **weight** mg downward and the **drag** D upward. The resultant force (Newton's second law, which you meet in the next topic) gives the acceleration:

$$\mathrm{a} = (\mathrm{mg} - \mathrm{D}) \,/\, \mathrm{m} \tag{1.22}$$

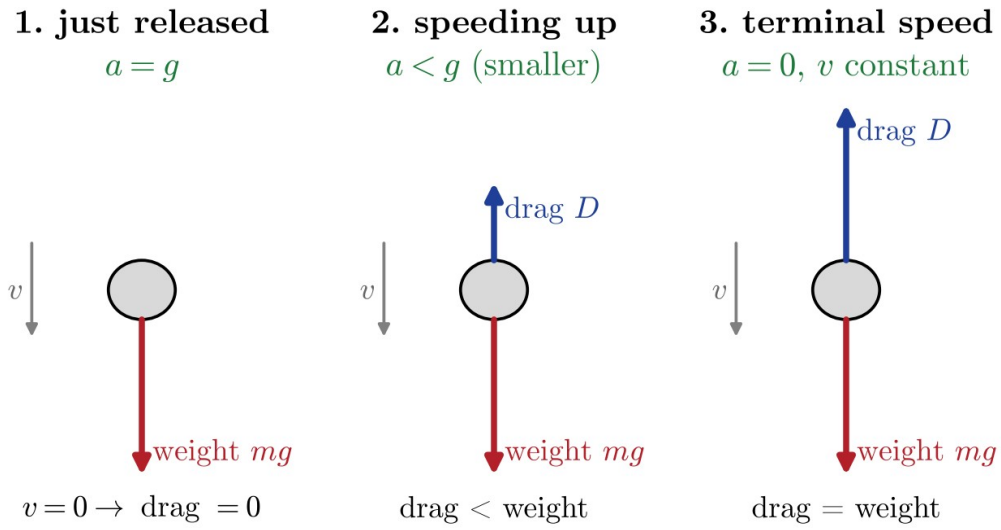


Figure 1.15: Forces on a falling object at three stages. As it speeds up, the upward drag grows until it equals the weight.

Follow the story in Figure 1.15 step by step. This chain of reasoning is exactly what exam questions want to see:

1. Released from rest: speed is zero, so drag is zero. The only force is weight, so **a** = **g**.
2. As the speed increases, drag **increases**. The resultant downward force **decreases**, so the acceleration **decreases** (but the object is still speeding up).
3. Eventually drag **equals** the weight. The resultant force is **zero**, so a = 0 and the velocity is constant. This is the **terminal speed**.

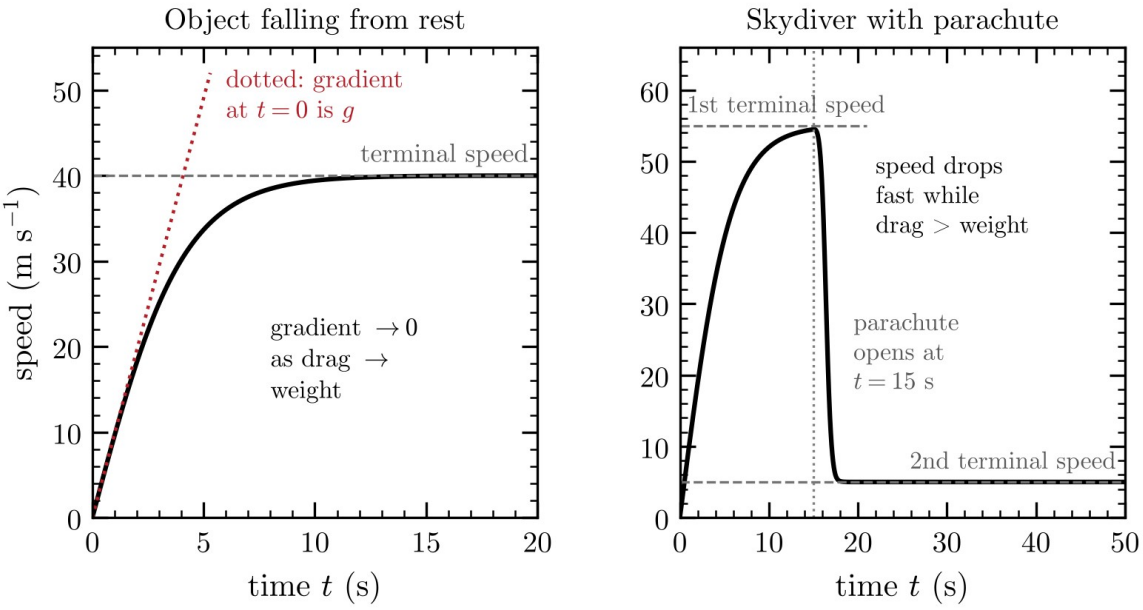


Figure 1.16: Left: speed against time for an object falling from rest; the gradient starts at g and falls to zero as the speed approaches its terminal value. Right: a skydiver. Opening the parachute suddenly makes drag larger than weight, so the skydiver slows down to a new, much lower, terminal speed.

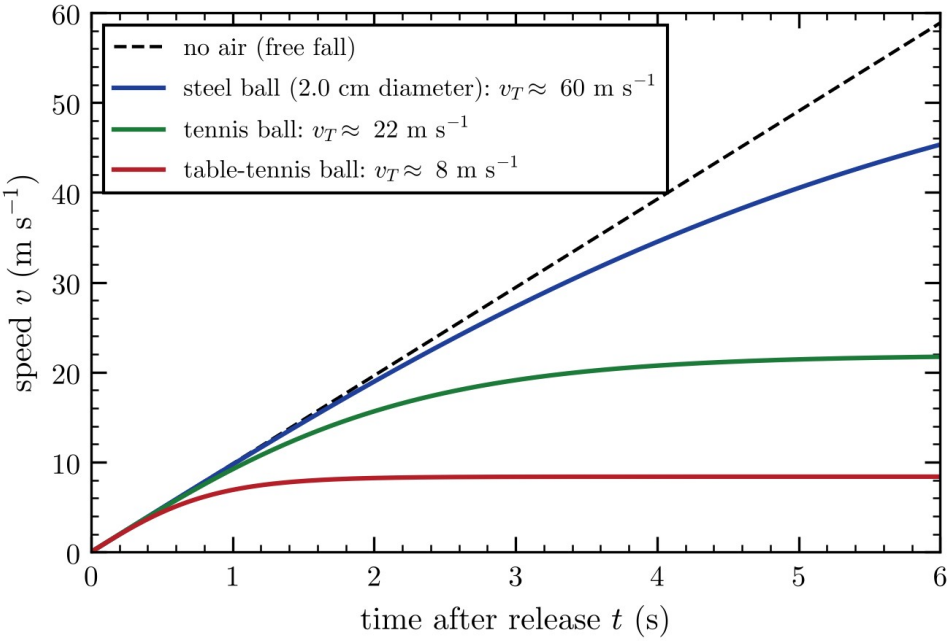


Figure 1.17: Computed speed–time graphs for three balls dropped from rest in air, compared with free fall (no air). The light table-tennis ball reaches its low terminal speed in about a second; the small, dense steel ball is still close to free fall after several seconds.

KEY IDEA **Terminal speed** is the constant speed an object reaches when the drag force equals its weight (resultant force zero, acceleration zero). A parachute has a huge area, so it needs a much *lower* speed before drag can balance the weight.

WATCH OUT Terminal speed does **not** mean the object has stopped accelerating *because gravity switched off*. Gravity is still acting; it is simply balanced by the drag. Also note that the speed-time graph has a gradient which is *decreasing*, not that the speed is decreasing.

Drag on a thrown object

When an object is thrown **upward** through the air, both weight and drag point **down**, so on the way up the acceleration is *greater* than g. At the top the speed is momentarily zero, so drag is zero and the acceleration is exactly g. On the way **down** drag points up, so the acceleration is *less* than g.

Projectiles with fluid resistance (qualitative)

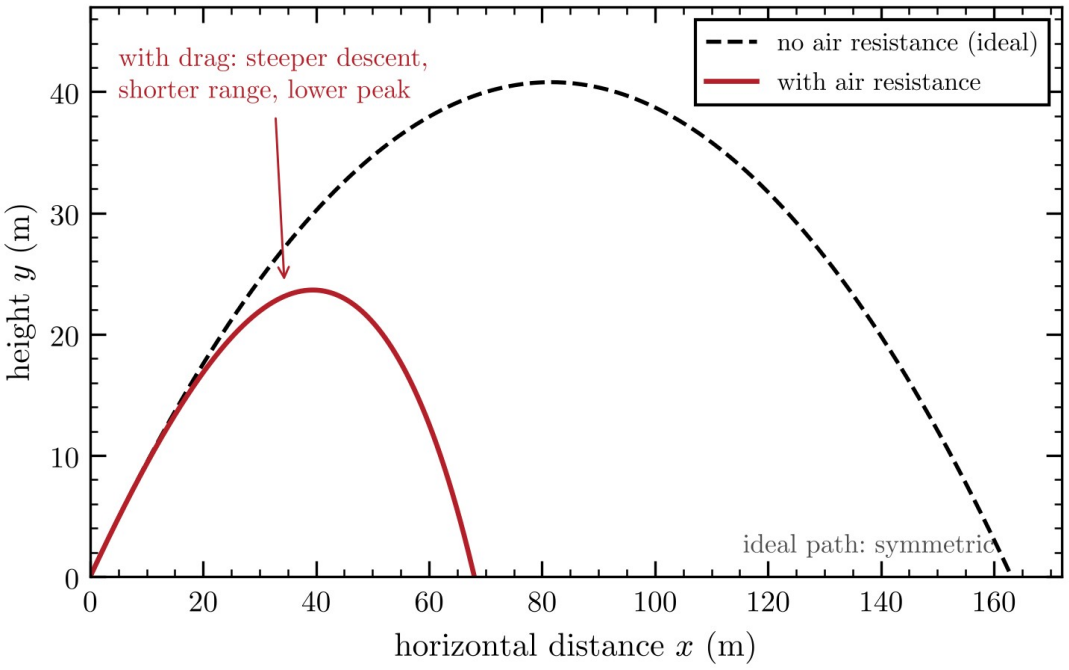


Figure 1.18: A projectile launched at 40 m s^{-1} at 45° with and without air resistance (computer model). Drag makes the path lopsided: lower, shorter, and steeper on the way down.

Quantity	**No air resistance**	**With air resistance**
Horizontal velocity	constant	**decreases** all the way (drag acts backwards)
Vertical acceleration	always $-g$	larger than g on the way up; exactly g at the top; smaller than g on the way down
Maximum height	H	**lower**
Range	R	**shorter**
Time of flight	$2u \sin\vartheta / g$	usually **shorter**; the descent takes longer than the ascent
Shape of path	symmetric parabola	**asymmetric**, steeper on the way down
Landing angle	same as launch angle	**steeper** than the launch angle
Landing speed	equal to launch speed	**smaller** than launch speed

1.11 Experiments: measuring g and uncertainty

A classic experiment measures g by dropping a small, dense object from different heights and timing the fall. Starting from rest, $s = \frac{1}{2}gt^2$ (Eq. 1.8 with $u = 0$ and $a = g$). This is not a straight line in s against t, but it **becomes** one if you plot s against $\mathbf{t^2}$:

$$s = (g/2) \times t^2 \quad \text{so} \quad g = 2 \times (\text{gradient of the } s\text{–}t^2 \text{ graph}) \tag{1.23}$$

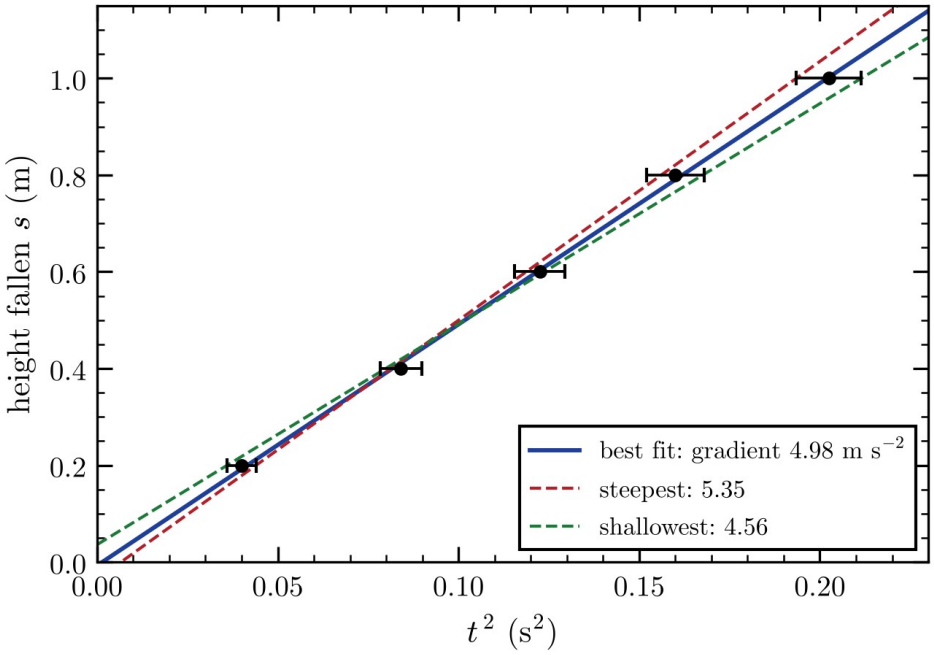


Figure 1.19: Height fallen against t^2 for a dropped ball, with error bars in t^2. The best-fit line (blue) and the steepest and shallowest lines (dashed) that still pass through all the error bars give the uncertainty in the gradient. The data are used in Example 21.

Every measurement has an **uncertainty**. Use these rules to combine them:

Operation	**Rule for the uncertainty**	**Example**
$a \pm b$	**add the absolute uncertainties**	$(5.0 \pm 0.1) + (3.0 \pm 0.1) = 8.0 \pm 0.2$
$a \times b$ or $a \div b$	**add the percentage (fractional) uncertainties**	$10 \pm 2\%$ and $5 \pm 3\% \rightarrow$ product $\pm 5\%$
a^n	**multiply the percentage uncertainty by n**	$t = 0.50 \pm 2\% \rightarrow t^2$ has $\pm 4\%$
Gradient of a best-fit line	(steepest − shallowest) ÷ 2	$(5.35 - 4.56) \div 2 \approx 0.4$

Type of error	**Cause (example)**	**Effect and fix**
Random	reaction time when using a stopwatch; scatter in repeat readings	scatter of points; **repeat and average**
Systematic	zero error on the ruler; air resistance on the falling ball; a timer that starts late	all values shifted the same way; **cannot** be averaged away; **improve the method**

EXAM TIP Report the final answer with a sensible number of significant figures and its uncertainty, e.g. $\mathbf{g = 9.6 \pm 0.5\ m\ s^{-2}}$. The uncertainty is normally given to **one** significant figure, and the answer to the **same decimal place**.

1.12 Relative velocity in a straight line (enrichment)

Velocity always depends on who is measuring. A passenger walking at 1 m s^{-1} along a train moving at 20 m s^{-1} has a velocity of 21 m s^{-1} *relative to the ground* but 1 m s^{-1} relative to the train. For motion along a line (adding signed velocities):

$$v_{\text{A relative to C}} = v_{\text{A relative to B}} + v_{\text{B relative to C}} \tag{1.24}$$

A common case is a boat in a river. The boat's speed relative to the **water** is fixed by its engine; the water moves relative to the **ground**; the ground-based observer sees the sum.

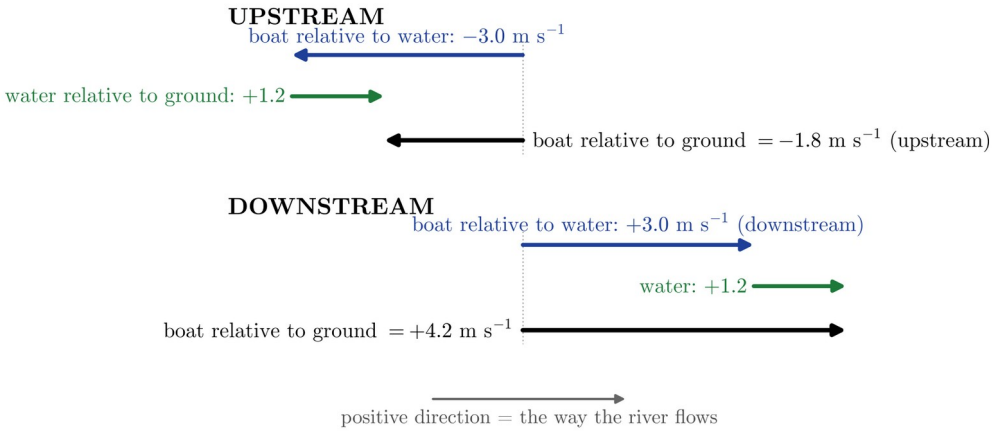


Figure 1.20: A boat that can move at 3.0 m s^{-1} relative to the water, in a river flowing at 1.2 m s^{-1}. Upstream the velocities partly cancel; downstream they add.

EXAM TIP For relative-velocity questions, choose a positive direction first, label each velocity with *"of what, relative to what"* (for example $v_{\text{boat/water}}$, $v_{\text{water/ground}}$) and then simply add. Check that the "inner" labels cancel: boat/water + water/ground = boat/ground.

Chapter 2

Worked Examples

Every example uses the same six steps from the front of this guide: **Draw, List, Choose, Rearrange, Substitute, Check**. Try each question yourself first. Take g = 9.81 m s^{-2} and neglect air resistance unless told otherwise.

2.1 Distance, displacement, speed and velocity

EXAMPLE 1 A runner jogs 400 m east along a straight road in 80 s, then turns round and walks 100 m west in 50 s. For the whole trip, find (a) the distance travelled, (b) the displacement, (c) the average speed, (d) the average velocity.

Draw Take east as positive. Leg 1: $\Delta s_1 = +400$ m. Leg 2: $\Delta s_2 = -100$ m. Total time t = 80 s + 50 s = 130 s.

(a) Distance adds the *sizes* of each leg:

$$\text{distance} = 400 \text{ m} + 100 \text{ m} = 500 \text{ m}$$

(b) Displacement adds the *signed* legs:

$$\Delta s = (+400 \text{ m}) + (-100 \text{ m}) = +300 \text{ m}$$

(c) Average speed = distance ÷ time:

$$500 \text{ m} \div 130 \text{ s} = 3.8 \text{ m s}^{-1}$$

(d) Average velocity = displacement ÷ time:

$$300 \text{ m} \div 130 \text{ s} = +2.3 \text{ m s}^{-1}$$

Check The velocity (2.3) is smaller than the speed (3.8), as it must be when the runner turns back.

Answer (a) 500 m (b) 300 m east (c) 3.8 m s^{-1} (d) 2.3 m s^{-1} east.

EXAMPLE 2 A car travels the first half of a 120 km journey at 60 km h^{-1} and the second half at 90 km h^{-1}. Find its average speed for the whole journey (a) in km h^{-1} and (b) in m s^{-1}.

List Each half is 60 km. We want total distance ÷ total time, so we need the **time** for each half. (Not the average of 60 and 90!)

Times First half: $t_1 = 60 \text{ km} \div 60 \text{ km h}^{-1} = 1.00$ h. Second half: $t_2 = 60 \text{ km} \div 90 \text{ km h}^{-1} = 0.667$ h.

(a) Average speed:

$$120 \text{ km} \div (1.00 \text{ h} + 0.667 \text{ h}) = 120 \div 1.667 = 72 \text{ km h}^{-1}$$

(b) Convert by dividing by 3.6:

$$72 \text{ km h}^{-1} \div 3.6 = 20 \text{ m s}^{-1}$$

Check The answer is **not** 75 km h^{-1} (the simple average), because the car spends *more time* at the slower speed. It is closer to 60 than to 90. ✓

Answer (a) 72 km h^{-1} (b) 20 m s^{-1}.

2.2 Reading motion graphs

EXAMPLE 3 **Figure 2.1 (top) shows the displacement of a cyclist from home. (a) Find her velocity in each of the three stages. (b) Find the distance travelled and the displacement over the 12 s. (c) Find the average speed and the average velocity. (d) Sketch the velocity–time graph.**

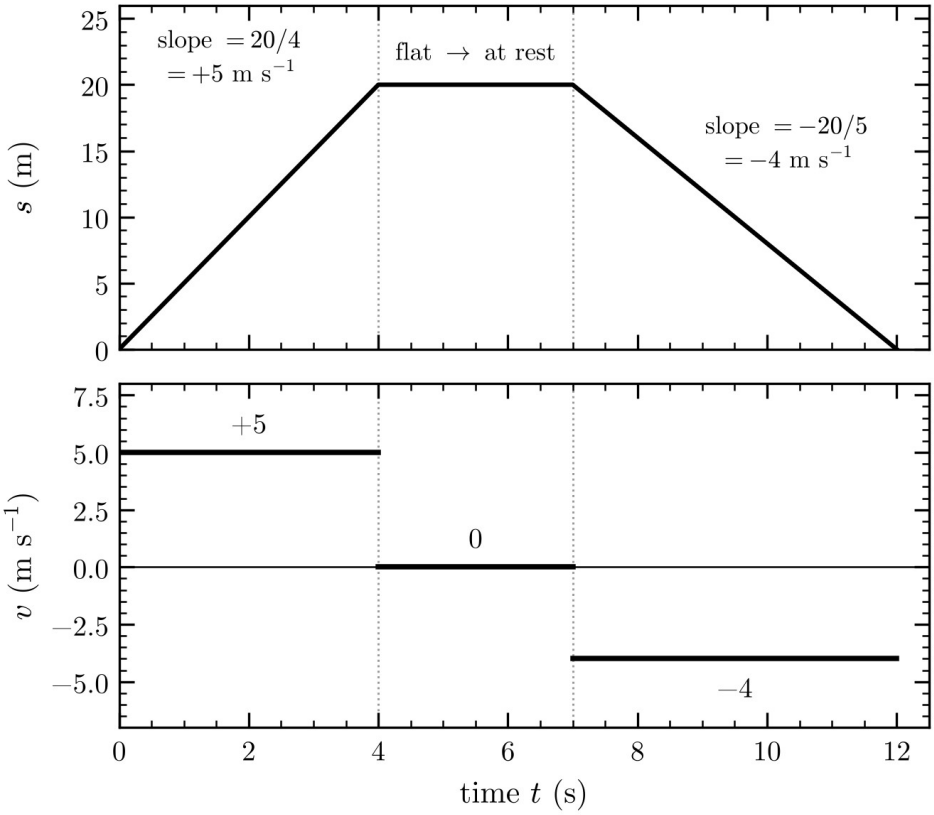


Figure 2.1: Top: displacement–time graph for Example 3. Bottom: the matching velocity–time graph, found from the gradients.

Rule On an s–t graph, **velocity = gradient**.

(a) Stage 1 (0–4 s): $(20 - 0) \div 4 = +\mathbf{5\ m\ s^{-1}}$. Stage 2 (4–7 s): flat line, so **0** (at rest). Stage 3 (7–12 s): $(0 - 20) \div 5 = -\mathbf{4\ m\ s^{-1}}$ (coming back home).

(b) Distance = 20 m out + 20 m back = **40 m**. Displacement = final − initial = 0 − 0 = **0** (she ends where she started).

(c) Average speed = 40 m ÷ 12 s = **3.3 m s^{-1}**. Average velocity = 0 ÷ 12 s = **0**.

(d) See the lower graph in Figure 2.1: three horizontal lines at +5, 0 and −4 m s^{-1}.

Check The area under the lower graph should equal the displacement: $+5 \times 4 = +20$ and $-4 \times 5 = -20$, which sum to 0. ✓

Answer (a) +5, 0, −4 m s^{-1} (b) 40 m and 0 (c) 3.3 m s^{-1} and 0.

EXAMPLE 4 The curve in Figure 1.3 shows $s = t^2$ (s in metres, t in seconds). (a) Find the average velocity between $t = 1.0$ s and $t = 4.0$ s. (b) Use a tangent to find the instantaneous velocity at $t = 3.0$ s. (c) Explain why the two values differ.

(a) A **chord** joins the points at t = 1 s (s = 1 m) and t = 4 s (s = 16 m):

$$\bar{v} = \Delta s \,/\, \Delta t = (16 - 1)\ \text{m} \div (4 - 1)\ \text{s} = 5.0\ \text{m s}^{-1}$$

(b) Draw the **tangent** at t = 3 s. Read two points on it, e.g. (2.0 s, 3.0 m) and (4.0 s, 15.0 m):

$$v = \text{rise} \div \text{run} = (15.0 - 3.0)\ \text{m} \div (4.0 - 2.0)\ \text{s} = 6.0\ \text{m s}^{-1}$$

(c) The curve gets steeper, so the velocity keeps **increasing** (tangents drawn at 1 s and 4 s give about 2 and 8 m s^{-1}). The average over 1–4 s is a mixture of all those values, while the tangent gives the value at one instant. Here 5.0 m s^{-1} is the velocity at the *middle* time (2.5 s), which is earlier than 3.0 s, so it is smaller.

Answer (a) 5.0 m s^{-1} (b) 6.0 m s^{-1} (c) the velocity is increasing; the average is not the value at t = 3 s.

EXAMPLE 5 Figure 2.2 shows the velocity of a trolley moving along a straight track (east is positive). (a) Find its acceleration in each stage. (b) At what time does it change direction? (c) Find the displacement and the distance travelled in the 16 s. (d) Find the average speed and the average velocity.

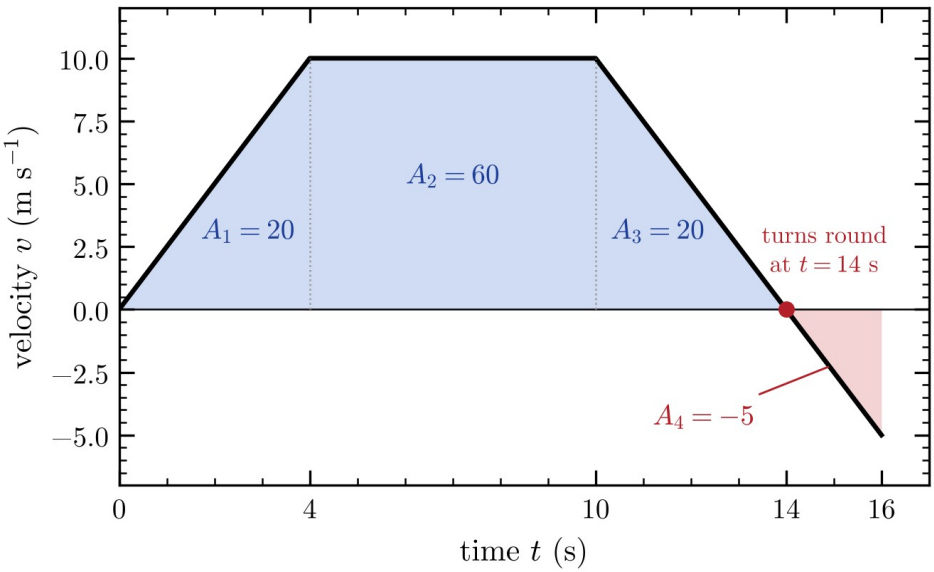


Figure 2.2: Velocity–time graph for Example 5. Areas above the axis are positive displacement; the area below is negative.

(a) Acceleration = gradient of v–t. 0–4 s: $(10 - 0) \div 4 = +\mathbf{2.5\ m\ s^{-2}}$. 4–10 s: **0**. 10–16 s: $(-5 - 10) \div 6 = -\mathbf{2.5\ m\ s^{-2}}$.

(b) The trolley turns round where v crosses zero. From t = 10 s the velocity falls by 2.5 m s^{-1} each second, so it reaches 0 after $10 \div 2.5 = 4$ s, i.e. at $\mathbf{t = 14\ s}$.

(c) Areas: $A_1 = \frac{1}{2} \times 4 \times 10 = 20$ m. $A_2 = 10 \times 6 = 60$ m. $A_3 = \frac{1}{2} \times 4 \times 10 = 20$ m. $A_4 = -\frac{1}{2} \times 2 \times 5 = -5$ m (below the axis).

$$\text{displacement} = 20 + 60 + 20 - 5 = 95\ \text{m}$$

$$\text{distance} = 20 + 60 + 20 + 5 = 105\ \text{m}$$

(d) Average speed = 105 ÷ 16 = **6.6 m s^{-1}**. Average velocity = 95 ÷ 16 = **5.9 m s^{-1}** (east).

Check The trolley is furthest from the start at t = 14 s (+100 m), then comes back 5 m, so the displacement of 95 m makes sense.

Answer (a) +2.5, 0, −2.5 m s^{-2} (b) 14 s (c) 95 m and 105 m (d) 6.6 m s^{-1} and 5.9 m s^{-1} east.

EXAMPLE 6 A trolley starts from rest. Its acceleration is shown in Figure 2.3 (top). (a) Find the velocity at t = 5, 8 and 12 s. (b) Sketch the velocity–time graph. (c) Find the total displacement in 12 s.

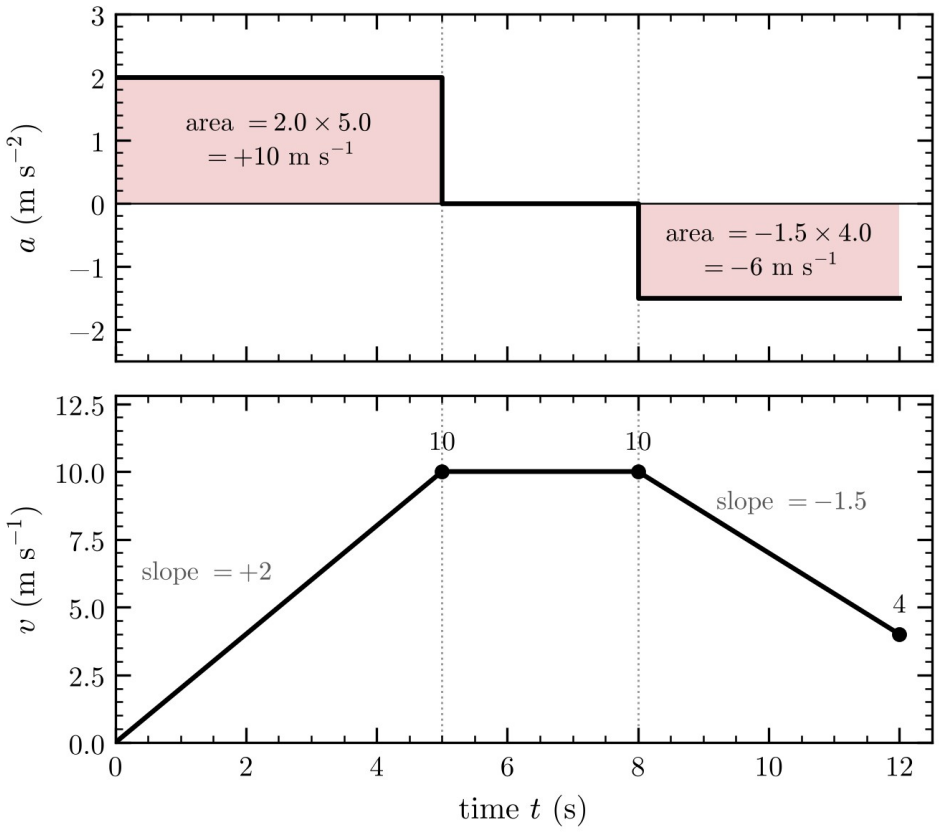


Figure 2.3: Top: acceleration–time graph for Example 6. Bottom: the velocity–time graph built from the areas.

Rule On an a–t graph, **area = change in velocity**. Start from v = 0.

(a) 0–5 s: area = 2.0 × 5.0 = +10, so v(5 s) = 0 + 10 = **10 m s^{-1}**. 5–8 s: a = 0, so v(8 s) = **10 m s^{-1}**. 8–12 s: area = −1.5 × 4.0 = −6, so v(12 s) = 10 − 6 = **4 m s^{-1}**.

(b) Join the points (0, 0), (5, 10), (8, 10), (12, 4) with straight lines: Figure 2.3 (bottom). The slopes are +2, 0 and −1.5, matching the a–t graph.

(c) Displacement = area under the v–t graph: triangle $\frac{1}{2} \times 5 \times 10 = 25$ m; rectangle $3 \times 10 = 30$ m; trapezium $\frac{1}{2} \times (10 + 4) \times 4 = 28$ m.

$$s = 25 + 30 + 28 = 83 \text{ m}$$

Answer (a) 10, 10, 4 m s^{-1} (c) 83 m.

2.3 Constant acceleration (SUVAT)

EXAMPLE 7 A car travelling at 25 m s^{-1} brakes with constant deceleration and stops in a distance of 50 m. (a) Find the deceleration. (b) Find the stopping time. (c) How far would it need to stop from 50 m s^{-1} with the same braking?

Draw Take the direction of travel as positive. u = +25 m s^{-1}, v = 0, s = +50 m. Unknown: a (and t). Time is not given, so choose the equation with **no t**.

(a) $v^2 = u^2 + 2as$, so $a = (v^2 - u^2) \div 2s$:

$$a = (0 - 25^2) \div (2 \times 50) = -625 \div 100 = -6.25 \text{ m s}^{-2}$$

(b) $v = u + at$, so $t = (v - u) \div a$:

$$t = (0 - 25) \div (-6.25) = 4.0 \text{ s}$$

(c) Same a, new u = 50 m s^{-1}:

$$s = (0 - 50^2) \div (2 \times (-6.25)) = 200 \text{ m}$$

Check Use $s = \frac{1}{2}(u + v)t$ for part (a)–(b): $\frac{1}{2} \times 25 \times 4.0 = 50$ m ✓. In (c), doubling the speed made the stopping distance **four** times bigger, since $s \propto u^2$ (this is why speed limits matter).

Answer (a) 6.25 m s^{-2} backwards ($a = -6.25$ m s^{-2}) (b) 4.0 s (c) 200 m.

EXAMPLE 8 A train accelerates from rest at 0.50 m s^{-2} for 40 s, then travels at constant speed for 120 s, then brakes uniformly and stops 20 s later. Find (a) the maximum speed, (b) the deceleration, (c) the total distance, (d) the average speed for the whole journey.

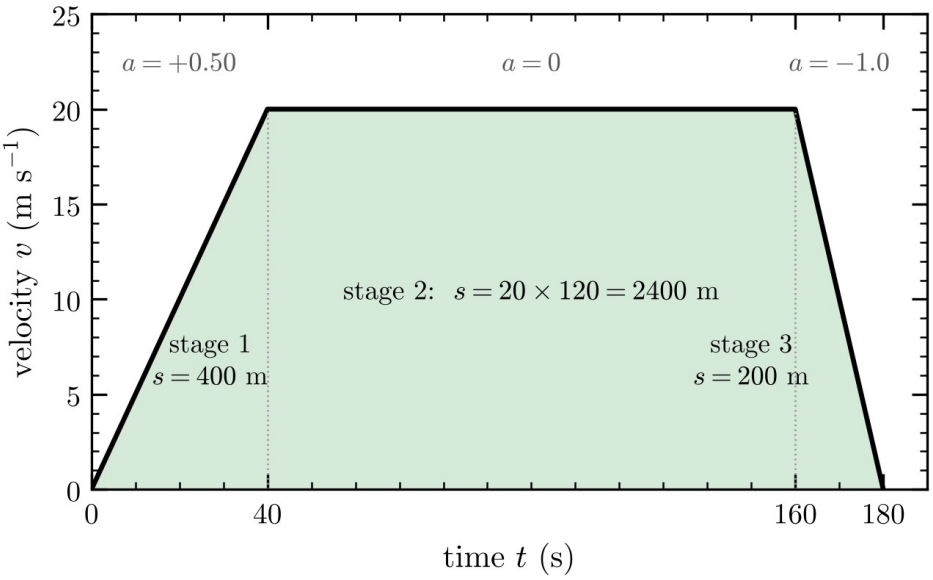


Figure 2.4: Velocity–time graph for the train in Example 8. The total area is the total distance.

Draw The motion has **three stages** with different accelerations, so SUVAT must be applied to each stage separately. The final speed of stage 1 is the speed of stage 2, and so on.

(a) Stage 1: u = 0, a = 0.50, t = 40 s. $v = u + at$:

$$v = 0 + 0.50 \times 40 = 20 \text{ m s}^{-1}$$

(b) Stage 3: u = 20, v = 0, t = 20 s:

$$a = (0 - 20) \div 20 = -1.0 \text{ m s}^{-2}$$

(c) Stage 1: $s_1 = \frac{1}{2}(u + v)t = \frac{1}{2} \times 20 \times 40 = 400$ m. Stage 2: $s_2 = 20 \times 120 = 2400$ m. Stage 3: $s_3 = \frac{1}{2} \times 20 \times 20 = 200$ m.

$$\text{total distance} = 400 + 2400 + 200 = 3000 \text{ m}$$

(d) Total time $= 40 + 120 + 20 = 180$ s.

$$\text{average speed} = 3000 \div 180 = 17 \text{ m s}^{-1}$$

Check The area under Figure 2.4 is a trapezium: $\frac{1}{2} \times (120 + 180) \times 20 = 3000$ m ✓. The average speed (17) lies below the maximum speed (20) as it should.

Answer (a) 20 m s^{-1} (b) -1.0 m s^{-2} (c) 3000 m (d) 17 m s^{-1}.

EXAMPLE 9 A car travelling at a constant 20 m s^{-1} passes a stationary police car. At that moment the police car accelerates from rest at 4.0 m s^{-2} in the same direction. (a) When does the police car catch up? (b) How far have they travelled? (c) How fast is the police car going then?

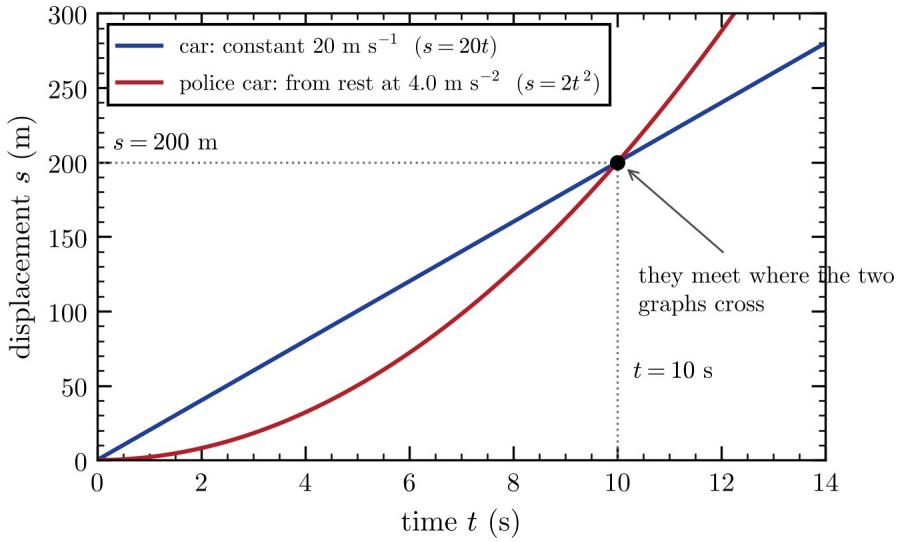


Figure 2.5: Displacement–time graphs for the two cars in Example 9. They are at the same place when the graphs cross.

Draw Set $t = 0$ and $s = 0$ when the car passes the police car. "Catching up" means **equal displacements at the same time**. Write one equation for each car.

Car $a = 0$, so $s_{car} = vt = 20t$.

Police $u = 0$, $a = 4.0$, so $s_{police} = \frac{1}{2}at^2 = 2t^2$.

(a) Set them equal:

$$2t^2 = 20t \quad \rightarrow \quad 2t(t - 10) = 0 \quad \rightarrow \quad t = 0 \text{ or } t = 10 \text{ s}$$

The root $t = 0$ is the starting moment (the cars were side by side then). The catch-up is at $\mathbf{t = 10\ s}$.

(b) $s = 20 \times 10 = \mathbf{200\ m}$.

(c) $v = u + at = 0 + 4.0 \times 10 = \mathbf{40\ m\ s^{-1}}$.

Check The police car's *average* speed over the 10 s was $\frac{1}{2}(0 + 40) = 20$ m s^{-1}, the same as the car's: that is why they cover equal distances. Its final speed is **double** the car's. ✓

Answer (a) 10 s (b) 200 m (c) 40 m s^{-1}.

2.4 Free fall

EXAMPLE 10 A ball is thrown vertically upward from the ground at 18 m s^{-1}. Find (a) the time to reach its highest point, (b) the maximum height, (c) the total time in the air, (d) its velocity and height 3.0 s after it is thrown.

Draw Up is positive, origin at the ground. $u = +18$ m s^{-1}, $a = -9.81$ m s^{-2} for the whole flight.

(a) At the top $v = 0$. $v = u + at$ gives:

$$t = (0 - 18) \div (-9.81) = 1.83 \text{ s}$$

(b) $v^2 = u^2 + 2as$ with $v = 0$:

$$s = (0 - 18^2) \div (2 \times (-9.81)) = 16.5 \text{ m}$$

(c) Up and down take equal times:

$$T = 2 \times 1.83 = 3.67 \text{ s}$$

(d) $v = u + at = 18 - 9.81 \times 3.0 = -11.4$ m s^{-1}. Height: $s = ut + \frac{1}{2}at^2 = 18 \times 3.0 - \frac{1}{2} \times 9.81 \times 3.0^2 = 54 - 44.1 = 9.9$ m.

Check 3.0 s is *after* the top (1.83 s) but *before* landing (3.67 s). So the ball should be moving **downward** (v negative ✓) and still above the ground ✓.

Answer (a) 1.83 s (b) 16.5 m (c) 3.67 s (d) 11.4 m s^{-1} downward, at a height of 9.9 m.

EXAMPLE 11 A ball is thrown vertically upward at 12 m s^{-1} from the top of a 30 m tall building. Find (a) the time to reach the ground and (b) its speed on landing.

Draw Up is positive, origin at the launch point (top of the building). $u = +12$ m s^{-1}, $a = -9.81$ m s^{-2}, and the ground is **below** the origin: $s = -30$ m.

(a) No v given, and we want t, so use $s = ut + \frac{1}{2}at^2$:

$$-30 = 12t - 4.905t^2 \quad \rightarrow \quad 4.905t^2 - 12t - 30 = 0$$

$$t = [12 \pm \sqrt{(12^2 + 4 \times 4.905 \times 30)}] \div (2 \times 4.905) = (12 \pm 27.07) \div 9.81$$

This gives $t = 3.98$ s or $t = -1.54$ s. A negative time is before the throw, so we **reject** it: $t = 3.98$ s.

(b) Time is not needed here, so use $v^2 = u^2 + 2as$:

$$v^2 = 12^2 + 2 \times (-9.81) \times (-30) = 144 + 588.6 = 732.6 \quad \rightarrow \quad v = -27.1 \text{ m s}^{-1}$$

Check Using $v = u + at$: $12 - 9.81 \times 3.98 = -27.1$ m s^{-1} ✓. The negative sign means downward.

Answer (a) 3.98 s (b) 27 m s^{-1} downward.

EXAMPLE 12 A stone is dropped from rest from a bridge. It falls the last 20 m in exactly 1.0 s before hitting the water. How high is the bridge above the water?

Draw Take down as positive for this problem, so $a = +9.81$ m s^{-2}. Split the fall into two parts: (i) from rest to the start of the last second, (ii) the last second. The speed v_1 at the junction is the link.

Part (ii) Known: $s = 20$ m, $t = 1.0$ s, $a = 9.81$, unknown $u = v_1$. Use $s = ut + \frac{1}{2}at^2$:

$$20 = v_1 \times 1.0 + \tfrac{1}{2} \times 9.81 \times 1.0^2 \quad \rightarrow \quad v_1 = 20 - 4.905 = 15.1 \text{ m s}^{-1}$$

Part (i) From rest to v_1: $v_1^2 = 2gh_1$, so

$$h_1 = 15.1^2 \div (2 \times 9.81) = 11.6 \text{ m}$$

Total height $= 11.6$ m $+ 20$ m $= 31.6$ m.

Check Final speed $= 15.1 + 9.81 \times 1.0 = 24.9$ m s^{-1}. Then $h = v^2 \div 2g = 24.9^2 \div 19.62 = 31.6$ m ✓.

Answer The bridge is 32 m above the water (31.6 m).

2.5 Projectile motion

EXAMPLE 13 **A ball rolls off the edge of a table 1.20 m high with a horizontal speed of 2.5 m s^{-1}. Find (a) the time to hit the floor, (b) how far from the table it lands, (c) the speed and direction of its velocity just before impact.**

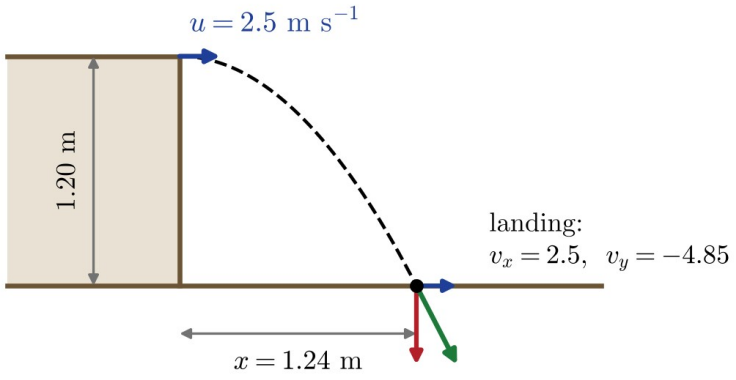


Figure 2.6: The ball in Example 13 leaves the table horizontally.

Draw Origin at the table edge, up positive. Launched horizontally: $u_x = 2.5$ m s^{-1}, $u_y = 0$. The floor is at $s_y = -1.20$ m. Vertical: $a = -9.81$. Horizontal: $a = 0$.

(a) Vertical motion gives the time. $s_y = u_y t - \frac{1}{2}gt^2$ with $u_y = 0$:

$$-1.20 = -\tfrac{1}{2} \times 9.81 \times t^2 \quad \rightarrow \quad t^2 = 0.2446 \quad \rightarrow \quad t = 0.495 \text{ s}$$

(b) Horizontal motion uses that time:

$$s_x = u_x t = 2.5 \times 0.495 = 1.24 \text{ m}$$

(c) $v_x = 2.5$ m s^{-1} (unchanged). $v_y = u_y - gt = -9.81 \times 0.495 = -4.85$ m s^{-1}.

$$v = \sqrt{(2.5^2 + 4.85^2)} = 5.46 \text{ m s}^{-1} \qquad \tan\vartheta = 4.85 \div 2.5 \quad \rightarrow \quad \vartheta = 62.7^\circ$$

Check The speed (5.46) is bigger than the launch speed (2.5) because the ball gained speed falling. The direction is steeply downward (62.7° below horizontal), as expected after a 1.2 m fall. ✓

Answer (a) 0.495 s (b) 1.24 m (c) 5.5 m s^{-1} at 63° below the horizontal.

EXAMPLE 14 A golf ball is hit from level ground at 28 m s^{-1} at 35° above the horizontal. Find (a) the time of flight, (b) the maximum height, (c) the range, (d) the speed at the top of the flight.

Resolve $u_x = 28\cos 35° = 22.9$ m s^{-1}. $u_y = 28\sin 35° = 16.1$ m s^{-1}.

(a) Landing at the launch height, so $s_y = 0$ (or use $T = 2u_y/g$):

$$T = 2 \times 16.06 \div 9.81 = 3.27 \text{ s}$$

(b) At the top $v_y = 0$, so $v_y^2 = u_y^2 - 2gH$:

$$H = 16.06^2 \div (2 \times 9.81) = 13.1 \text{ m}$$

(c) Horizontal motion at constant speed:

$$R = u_x\, T = 22.94 \times 3.274 = 75.1 \text{ m}$$

(d) At the top $v_y = 0$, so the speed is just the horizontal component: **22.9 m s^{-1}**.

Check The shortcut $R = u^2 \sin 2\vartheta \div g = 28^2 \times \sin 70° \div 9.81 = 75.1$ m ✓ (allowed here because the launch and landing heights are equal and there is no air resistance).

Answer (a) 3.27 s (b) 13.1 m (c) 75.1 m (d) 22.9 m s^{-1}.

EXAMPLE 15 A ball is thrown at 18 m s^{-1} at 35° above the horizontal towards a wall 20 m away (Figure 2.7). The wall is 3.0 m high. (a) Does the ball clear the wall? (b) Is the ball still rising or already falling when it reaches the wall?

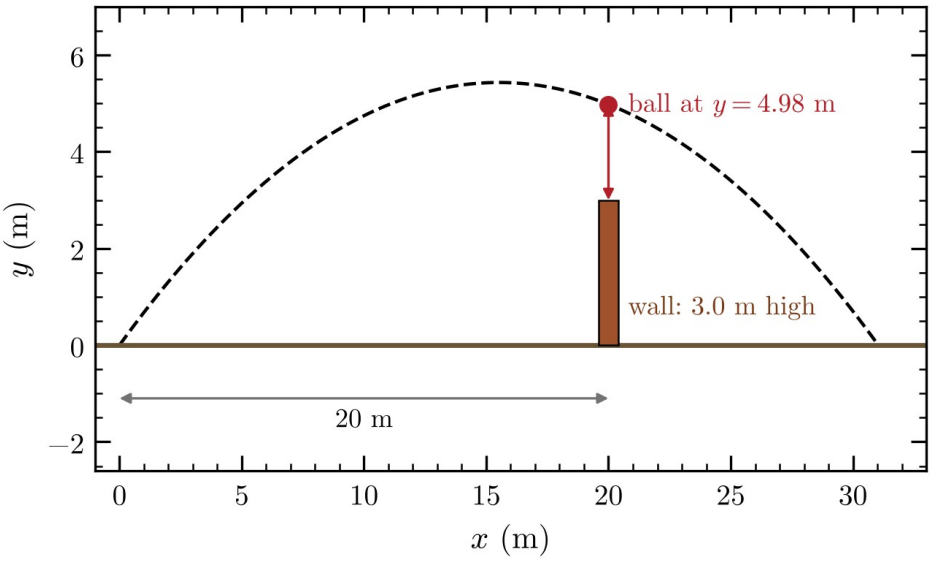


Figure 2.7: Example 15. The ball passes above the wall; its height at x = 20 m is found from the vertical equation.

Resolve $u_x = 18\cos 35° = 14.74$ m s^{-1}. $u_y = 18\sin 35° = 10.32$ m s^{-1}. The ball starts at ground level.

Time The wall is at $s_x = 20$ m, so (horizontal motion):

$$t = 20 \div 14.74 = 1.36 \text{ s}$$

(a) Height at that time: $s_y = u_y\, t - \tfrac{1}{2}gt^2$

$$s_y = 10.32 \times 1.356 - \tfrac{1}{2} \times 9.81 \times 1.356^2 = 14.00 - 9.03 = 4.98 \text{ m}$$

4.98 m > 3.0 m, so the ball **clears** the wall, by about 2.0 m.

(b) $v_y = u_y - gt = 10.32 - 9.81 \times 1.356 = \mathbf{-2.98\ m\ s^{-1}}$. This is negative, so the ball is already **falling**: it has passed its highest point.

Check The time to the top is $u_y/g = 1.05$ s, which is less than 1.36 s ✓.

Answer (a) Yes, at a height of 4.98 m (1.98 m above the wall). (b) Falling ($v_y = -2.98$ m s^{-1}).

EXAMPLE 16 A stone is thrown from the top of a 20 m high cliff at 15 m s^{-1}, 30° above the horizontal (Figure 2.8). Find (a) the time to reach the sea, (b) the horizontal distance from the base of the cliff, (c) the speed and direction of the stone as it hits the water.

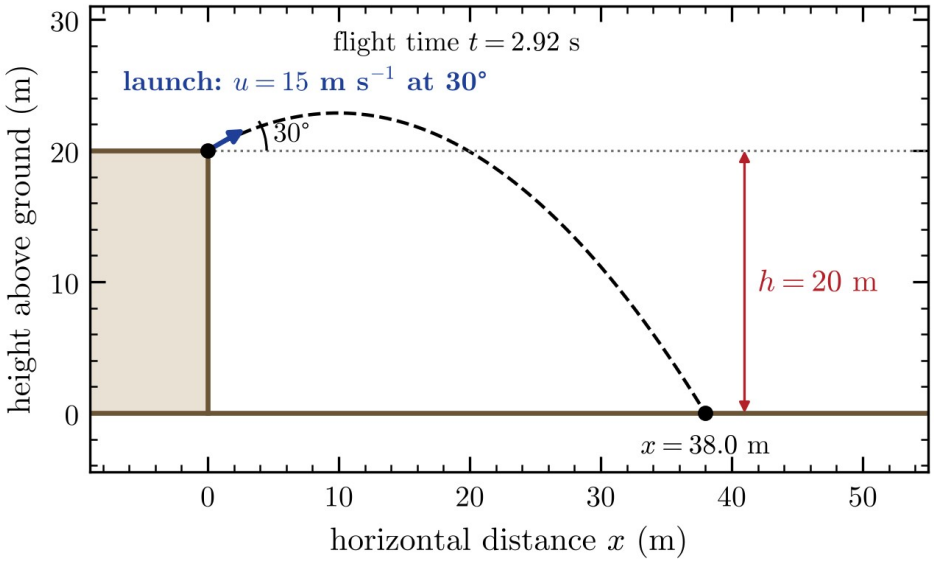


Figure 2.8: Example 16. The stone lands 20 m *below* its launch point, so the same-level shortcuts cannot be used.

Draw Origin at the launch point, up positive. The sea is **below**: $s_y = -20$ m. $u_x = 15\cos 30° = 12.99$ m s^{-1}, $u_y = 15\sin 30° = 7.50$ m s^{-1}.

(a) Vertical: $s_y = u_y\,t - \frac{1}{2}gt^2$

$$-20 = 7.50t - 4.905t^2 \quad\rightarrow\quad 4.905t^2 - 7.50t - 20 = 0$$

$$t = [7.50 \pm \sqrt{(7.50^2 + 4 \times 4.905 \times 20)}] \div 9.81 = (7.50 \pm 21.18) \div 9.81$$

$t = 2.92$ s or $t = -1.39$ s. Reject the negative root: $\mathbf{t = 2.92\ s}$.

(b) Horizontal: $s_x = u_x\,t = 12.99 \times 2.92 = \mathbf{38.0\ m}$.

(c) $v_x = 12.99$ m s^{-1}. $v_y = u_y - gt = 7.50 - 9.81 \times 2.92 = -21.2$ m s^{-1}.

$v = \sqrt{(12.99^2 + 21.2^2)} = 24.8$ m s^{-1} $\quad \tan\vartheta = 21.2 \div 12.99 \quad\rightarrow\quad \vartheta = 58.5°$ below horizontal

Check Speed from the no-time equation: $v^2 = u^2 + 2g \times 20 = 225 + 392.4 = 617.4$, so $v = 24.8$ m s^{-1} ✓.

Answer (a) 2.92 s (b) 38.0 m (c) 24.8 m s^{-1} at 58.5° below the horizontal.

EXAMPLE 17 A projectile is launched from level ground so that its horizontal range is eight times its maximum height. Find the launch angle.

Choose The launch and landing heights are equal, so the shortcuts $R = 2u^2 \sin\vartheta\cos\vartheta \div g$ and $H = u^2\sin^2\vartheta \div 2g$ are valid.

Rearrange Divide R by H:

$$R \div H = (2u^2 \sin\vartheta \cos\vartheta \div g) \div (u^2 \sin^2\vartheta \div 2g) = 4 \cos\vartheta \div \sin\vartheta = 4 \div \tan\vartheta$$

Solve $R = 8H$, so $4 \div \tan\vartheta = 8$, giving $\tan\vartheta = 0.50$:

$$\vartheta = \tan^{-1}(0.50) = 26.6^\circ$$

Check At $\vartheta = 45^\circ$ we get $R \div H = 4 \div 1 = 4$. A longer, flatter flight (larger $R \div H$) needs a smaller angle, so $26.6^\circ < 45^\circ$ is sensible. ✓ The launch speed cancelled out, so it is not needed.

Answer $\vartheta = 26.6^\circ$.

2.6 Fluid resistance and terminal speed

EXAMPLE 18 A skydiver of mass 80 kg falls with her arms and legs spread out. (a) What is the drag force on her when she is at terminal speed?

(b) At one moment her speed is 30 m s^{-1} and the drag force is 230 N. What is her acceleration? (c) She opens her parachute. Explain the motion that follows.

Draw Down is positive. Two forces: weight $W = mg$ downward, drag D upward. Weight: $W = 80 \times 9.81 = 785$ N.

(a) At terminal speed the velocity is constant, so $a = 0$ and the resultant force is zero:

$$D = mg = 785 \text{ N}$$

(b) Resultant force $= W - D = 785 - 230 = 555$ N downward, so

$$a = (mg - D) \div m = 555 \div 80 = 6.9 \text{ m s}^{-2} \text{ downward}$$

(c) Explain step by step:

- Opening the parachute suddenly increases the area, so the drag becomes **greater than the weight**.
- The resultant force is now **upward**, so the acceleration is upward: the skydiver **slows down** (the velocity is downward but the acceleration is upward).
- As the speed falls, the drag **decreases**, so the upward resultant force gets smaller.
- When drag again **equals** the weight, the resultant force is zero and she falls at a **new, lower terminal speed**.

Check In (b) the acceleration (6.9) is less than g (9.81) ✓, because drag is already reducing the resultant force, but she is still speeding up.

Answer (a) 785 N (b) 6.9 m s^{-2} downward (c) she slows down to a lower terminal speed (see steps above).

EXAMPLE 19 A ball is kicked at an angle from level ground. Describe and explain two ways in which its path would differ if air resistance were significant (compare with Figure 1.18).

Approach "Describe and explain" needs a **difference** and a **reason** for each. Link each reason to the drag force or to a velocity component.

- **Shorter range.** Drag acts opposite to the velocity, so it has a backwards horizontal component. The horizontal velocity therefore decreases continuously (instead of staying constant), and in the same time the ball travels less far horizontally.
- **Lower maximum height.** On the way up both weight and drag act downward, so the deceleration is larger than g. The vertical velocity reaches zero sooner and at a lower height.
- **Steeper descent (asymmetric path).** The horizontal velocity has become smaller while gravity keeps building up the vertical velocity, so the velocity vector makes a steeper angle with the horizontal when it lands. The landing speed is also smaller than the launch speed.

Answer Any two of: shorter range, lower maximum height, steeper landing angle / asymmetric path, with the reason for each.

2.7 Experiments and uncertainty

EXAMPLE 20 A student drops a ball from rest and measures the height $s = 1.20 \pm 0.01$ m and the fall time $t = 0.50 \pm 0.01$ s. Use $g = 2s \div t^2$ to find g with its uncertainty.

Value $g = 2s \div t^2 = 2 \times 1.20 \div 0.50^2 = 9.6 \text{ m s}^{-2}$.

Percentages s: $0.01 \div 1.20 = 0.83\ \%$. t: $0.01 \div 0.50 = 2.0\ \%$, and because t is **squared**, we double it: $4.0\ \%$.

Combine The quantities are multiplied/divided, so **add the percentage uncertainties**:

$$0.83\ \% + 4.0\ \% = 4.8\ \% \qquad \text{absolute: } 0.048 \times 9.6 = 0.46 \approx 0.5 \text{ m s}^{-2}$$

Check Notice that the timing is the bigger problem ($4.0\ \%$ against $0.83\ \%$). Using a more precise timer would improve the result the most. The accepted value 9.81 lies inside 9.6 ± 0.5. ✓

Answer $g = 9.6 \pm 0.5 \text{ m s}^{-2}$ (about 5 % uncertainty).

EXAMPLE 21 The table gives measurements for a ball dropped from rest. The uncertainty in each time is $\pm$ 0.01 s. Figure 1.19 shows the graph of s against t^2. Use the graph to find g and its uncertainty.

Height s (m)	Time t (s)	t^2 (s^2)	Uncertainty in t^2 ($\pm$ s^2)
0.20	0.20	0.0400	0.0040
0.40	0.29	0.0841	0.0058
0.60	0.35	0.1225	0.0070
0.80	0.40	0.1600	0.0080
1.00	0.45	0.2025	0.0090

Choose Starting from rest, $s = \frac{1}{2}gt^2$ is the equation of a straight line when s is plotted against t^2, with **gradient = g ÷ 2**.

Gradient From the best-fit line in Figure 1.19: gradient = 4.98 m s^{-2}. The steepest line through the error bars has gradient 5.35 and the shallowest has 4.56.

Uncertainty Uncertainty in the gradient = $(5.35 - 4.56) \div 2 = 0.40 \text{ m s}^{-2}$, so the gradient is $(4.98 \pm 0.40) \text{ m s}^{-2}$.

g $g = 2 \times$ gradient:

$$g = 2 \times (4.98 \pm 0.40) = 9.96 \pm 0.80 \text{ m s}^{-2} \approx 10.0 \pm 0.8 \text{ m s}^{-2}$$

Check The accepted value 9.81 lies within 10.0 ± 0.8 ✓. The line passes very close to the origin, which supports the assumption that the ball started from rest ($s = 0$ at $t = 0$). The uncertainty is about 8 %.

Answer $g = 10.0 \pm 0.8 \text{ m s}^{-2}$.

2.8 Relative velocity in a straight line (enrichment)

EXAMPLE 22 A boat can travel at 3.0 m s^{-1} in still water. It goes 100 m upstream and then 100 m back in a river that flows at 1.2 m s^{-1}. Find the total time and compare it with the time in still water.

Draw Take downstream as positive (Figure 1.20). Velocity of the water relative to the ground: +1.2 m s^{-1}.

Upstream Boat relative to water: −3.0. Boat relative to ground: $-3.0 + 1.2 = -1.8$ m s^{-1}, so the speed is 1.8 m s^{-1}. Time $= 100 \div 1.8 = 55.6$ s.

Downstream Boat relative to water: +3.0. Boat relative to ground: $+3.0 + 1.2 = +4.2$ m s^{-1}. Time $= 100 \div 4.2 = 23.8$ s.

$$\text{total time} = 55.6 + 23.8 = 79.4 \text{ s}$$

Still water Total distance 200 m at 3.0 m s^{-1}: $200 \div 3.0 = 66.7$ s.

Check The round trip with a current is **longer**. You lose more time going against the current than you gain going with it, because you spend more time at the slower speed. ✓

Answer 79 s with the current, compared with 67 s in still water.

Chapter 3

Quick Reference

3.1 Formula summary

Definitions

Quantity	Relationship	SI unit	Type
Displacement	$\Delta s = s_{final} - s_{initial}$	m	vector
Average speed	total distance ÷ total time	$m\ s^{-1}$	scalar
Average velocity	$\Delta s \div \Delta t$	$m\ s^{-1}$	vector
Acceleration	$a = (v - u) \div t$	$m\ s^{-2}$	vector
Unit conversion	$1\ m\ s^{-1} = 3.6\ km\ h^{-1}$	—	—
Components	$v_x = v \cos \vartheta, \quad v_y = v \sin \vartheta$	$m\ s^{-1}$	—
Magnitude and direction	$v = \sqrt{(v_x^2 + v_y^2)}, \quad \tan \vartheta = \lvert v_y \rvert \div \lvert v_x \rvert$	$m\ s^{-1}$, °	—

Constant acceleration (SUVAT)

Equation	Missing variable	Condition
$v = u + at$	s	constant acceleration, straight line
$s = \frac{1}{2}(u + v)t$	a	constant acceleration, straight line
$s = ut + \frac{1}{2}at^2$	v	constant acceleration, straight line
$v^2 = u^2 + 2as$	t	constant acceleration, straight line

Projectile motion (up positive, no air resistance)

Relationship	Meaning / condition
$u_x = u \cos \vartheta, \quad u_y = u \sin \vartheta$	ϑ measured from the horizontal
$v_x = u_x, \quad s_x = u_x\, t$	horizontal acceleration is zero
$v_y = u_y - gt$	vertical acceleration is $-g$
$s_y = u_y\, t - \frac{1}{2}gt^2$	vertical displacement from the launch point
$v_y^2 = u_y^2 - 2g\, s_y$	when time is not needed
$T = 2u \sin \vartheta \div g$	**only** if launch and landing heights are equal
$H = (u \sin \vartheta)^2 \div 2g$	**only** if launch and landing heights are equal
$R = u^2 \sin 2\vartheta \div g$	**only** if launch and landing heights are equal

Graph rules

Graph	Gradient	Signed area
s–t	velocity	—
v–t	acceleration	displacement
a–t	—	change in velocity

Fluid resistance and uncertainty

Idea	Statement
Falling with drag	$a = (mg - D) \div m$; at terminal speed $D = mg$, $a = 0$
Measuring g	plot s against t^2; $g = 2 \times$ gradient
$a \pm b$	add absolute uncertainties
$a \times b$, $a \div b$	add percentage uncertainties
a^n	multiply the percentage uncertainty by n
Gradient uncertainty	(steepest − shallowest) ÷ 2
Constant	$g = 9.81\ m\ s^{-2}$ (IB data booklet)

3.2 Which method? A decision guide

If the question...	Then...
gives a **graph**	gradient → rate of change (velocity from s–t, acceleration from v–t); area → displacement (v–t) or Δv (a–t). Say which axes you are reading.
says **constant** / **uniform acceleration**	write the SUVAT list (s, u, v, a, t), mark the knowns and the wanted quantity, choose the equation that leaves out the unused quantity.
has **two or more stages**	treat each stage separately. The final velocity of one stage is the initial velocity of the next. Or use a v–t graph and find areas.
is about a **ball thrown up** / **dropped**	free fall: $a = -g$ throughout (up positive). Remember $a \neq 0$ at the top.
has an object launched **at an angle**	resolve into components. Vertical gives the time; horizontal uses that time.
says the projectile **lands at a different height**	do **not** use T, H, R. Solve $s_y = u_y\, t - \frac{1}{2}gt^2$ with the right s_y (a quadratic), and keep the sensible root.
mentions **air resistance** / **drag** / **terminal speed**	use forces: weight down, drag up. $a = (mg - D)/m$. No SUVAT while the acceleration changes.
asks to **explain**	say what happens, then **why**, in terms of forces, components or the equation used.
involves **measurements with uncertainties**	apply the add / add-percent / multiply-percent rules, or use the steepest and shallowest lines on a graph.

FINAL CHECK Before you finish any question, ask four things: **Signs** consistent? **Units** written? **Direction** stated (if the quantity is a vector)? **Size** sensible (for example, a speed bigger than the speed of sound for a thrown ball is a red flag)?

3.3 Common mistakes and how to avoid them

Mistake	Why it is wrong	Fix
Using the average of two speeds as the average speed	time is spent at each speed in different amounts	total distance ÷ total time
Calling distance and displacement the same	displacement has direction and can be zero or negative	draw the path; add sizes for distance, signed values for displacement
Saying "negative acceleration means slowing down"	if v is also negative, it speeds up	compare the **directions** of v and a
Reading a v–t graph as if it were the path	the axes are velocity and time, not height and distance	say what each axis is first
Using SUVAT when a is not constant	the equations assume a constant acceleration	split into stages or use graphs
Putting $a = 0$ at the top of a throw	the velocity is still changing; only $v = 0$	$a = -g$ all the way
Mixing sign conventions in one problem	e.g. u positive but g also positive in an upward throw	choose once; apply to **s, u, v, a**
Using R, H, T for a cliff launch	they assume equal launch and landing heights	use the vertical equation and a quadratic
Mixing sin and cos	depends on whether the angle is from the horizontal or vertical	sketch the triangle; check the limits ($\vartheta \to 0$)
Keeping both roots of a quadratic	one root is usually a negative time	reject the physically impossible one
Saying terminal speed means "forces have stopped"	gravity and drag are still acting	say the forces are **balanced**: resultant force zero
Forgetting units or mixing km h^{-1} and m s^{-1}	answers are out by a factor of 3.6	convert at the start; keep units in every line

3.4 Command terms in kinematics questions

Command term	What it asks for	Short example (A.1)
State	a short answer, no working or reasoning	State the acceleration of a ball at the top of its flight. → 9.81 m s^{-2} downward.
Calculate	a numerical answer, with working, units and sensible significant figures	Calculate the range of a projectile...
Determine	find the answer; may use a graph or data	Determine g from the gradient.

Command term	**What it asks for**	**Short example (A.1)**
Sketch	a clear, labelled shape of the graph, not necessarily to scale	Sketch the v–t graph for the bouncing ball.
Explain	reasons, linked to physics ideas (forces, components, equations)	Explain why a skydiver reaches terminal speed.
Describe	what happens, with detail, in words	Describe the motion shown by the graph.
Derive	a step-by-step proof from stated assumptions	Derive $s = ut + \frac{1}{2}at^2$ for constant acceleration.
Discuss	a balanced account, with several points	Discuss the effect of air resistance on a projectile.

3.5 Self-check list

Tick a box only when you can explain the skill **and** apply it to a question you have not seen before.

✔	**I can...**	**See**
☐	distinguish scalars from vectors, and use signs for direction	1.2
☐	distinguish distance from displacement, and speed from velocity	1.2–1.3
☐	calculate average speed, average velocity and acceleration	1.3–1.4
☐	explain the difference between average and instantaneous values (chord and tangent)	1.3
☐	decide whether an object is speeding up or slowing down from the signs of v and a	1.4
☐	find velocity from an s–t gradient, and acceleration from a v–t gradient	1.5
☐	find displacement and distance from the areas on a v–t graph	1.5
☐	sketch and link s–t, v–t and a–t graphs for the same motion	1.5
☐	derive the SUVAT equations from a v–t graph	1.6
☐	choose the right SUVAT equation, and state its condition	1.6
☐	set a sign convention for vertical motion and keep it	1.7
☐	explain why $a \neq 0$ at the top of a throw	1.7
☐	resolve a vector into components, including when the angle is from the vertical	1.8
☐	solve a projectile problem using a shared time for horizontal and vertical motion	1.9
☐	use T, H and R only when they are valid	1.9
☐	solve a projectile problem that lands at a different height (quadratic, choose the root)	1.9
☐	explain terminal speed using forces	1.10
☐	describe how drag changes the path of a projectile	1.10
☐	determine g from a graph and propagate uncertainty	1.11
☐	use relative velocity along a line (enrichment)	1.12

Always check your own copy of the IB Physics guide and your teacher's advice for the exact scope of the course. Content marked "enrichment" goes beyond the core A.1 requirements.